

# EBB: Pooling under Forgetting for Probabilistic Intermittent-Demand Forecasting

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## Abstract

Intermittent-demand forecasters must decide how to pool evidence across items and how much history to retain. These choices are usually specified separately, although discounting changes both an item’s reliance on a shared prior and the information available to estimate that prior. We introduce EBB, a scalable hierarchical empirical-Bayes forecaster that applies one exponential recency operator to item- and group-level sufficient statistics and selects the discount jointly with the pooling partition by chronological validation. Forgetting has two opposing effects: it preserves prior leverage while reducing the effective sample available to resolve group heterogeneity. A two-sided credibility bound makes the trade-off explicit, and its resolution condition is estimable from the fitting window *before* any partition is fitted. Across five public panels comprising more than 19,000 series, under both a fixed-origin and a strict walk-forward protocol, EBB and an official Tweedie–Gaussian-process specialist are the only two methods within five percent of the best competing method on every panel. EBB leads on Auto and RAF under both protocols; the specialist leads on the remaining panels by under one percent in four of its five leads and by 2.9% on M5 at fixed origin, and it costs one to two orders of magnitude more and must be refit at every update, where EBB updates sufficient statistics in closed form. Adding the same adaptive conformal layer to EBB that the strongest conformal baseline uses makes it the walk-forward leader on M5 and Auto and closes Carparts to 0.2%, while hurting on Online Retail and RAF; a validation-tail rule inside the initialization window decides where to apply it, is right on four of five panels, and leaves EBB within 0.8% of the best competing method everywhere under walk-forward updating. The margins are modest, and the diagnostic explains what bounds them: the credibility weight an item retains after discounting caps how far any shared prior can move its posterior. Placing each panel on a controlled surface that varies leverage and group separation together, the study assigns room to one panel: Carparts, where a supplied true partition

earns 1.7 to 2.0% and the learned partition returns  $-0.08\%$ , a failure the same objective reproduces at the same coordinates in simulation. On the other four the surface predicts no room and none is found; across all five, refining the pooling structure improves external accuracy on none. The diagnostic thus separates panels without exploitable structure from the one panel where the structure exists and current mixture objectives do not recover it. Two evaluation artifacts recur throughout: absolute-error metrics reward the all-zero forecast on three panels, and the standard split-conformal wrapper understates the conformal baseline by roughly 20% relative to its adaptive counterpart. The contribution is therefore a cheap, deterministic predictive law together with an *ex ante* test of whether cross-series pooling is worth attempting at all.

## 1 Introduction

Intermittent demand, in which long runs of zeros are interrupted by sporadic positive quantities, is pervasive in spare-parts logistics, long-tail commerce, and supply chains. Forecasting many such series poses two coupled statistical choices. Evidence can be pooled across items to stabilize sparse estimates, but an overly coarse pool hides real heterogeneity. Old observations can be discounted to track drift and obsolescence, but forgetting also removes information. Local methods such as Croston, SBA, and TSB [5, 23, 25] emphasize temporal adaptation while estimating each item separately. Hierarchical probabilistic models [2, 18, 20] emphasize cross-series shrinkage, usually under a fixed pooling structure or a pre-specified temporal model. Pooling and forgetting are consequently often designed as separate decisions.

Figure 1 makes both choices concrete in real intermittent demand: the relevance of an item’s own history changes over time, while the histories from which it might borrow strength are sharply heterogeneous.

They are not separate statistically. With exponen-

tial forgetting, an item’s effective history remains bounded, so a shared prior can retain substantial influence even after a long observation sequence. The same discounting, however, reduces the effective information used to estimate each group-level prior. Stronger forgetting can therefore make pooling more consequential while making fine pooling structure harder to resolve. This tension is the paper’s organizing principle: the forgetting rate sets not only a temporal horizon, but also the statistical resolution at which cross-series structure can be learned.

We operationalize this principle with EBB,<sup>1</sup> a recency-weighted hierarchical empirical-Bayes forecaster for intermittent demand. The model uses an occurrence–size decomposition with Beta–Binomial occurrence and log-normal size blocks, represented by per-item sufficient statistics. The same exponential weights are applied to the statistics used for the item posterior and for group-level hyperparameter estimation. A small candidate family crosses discount factors with a global pool, the classical ADI/CV<sup>2</sup> taxonomy, and BIC-selected mixture partitions; chronological validation selects the pair before a full-window refit. For a fixed candidate family, estimation is deterministic and linear in the number of observations, with low-dimensional group optimization and mixture search.

The theory predicts an asymmetric empirical pattern. Forgetting should be a first-order forecasting decision, whereas refining an already pooled model can help only when the discounted data still resolve the proposed groups. Because the resolution condition depends on quantities available from the fitting window, it can be checked panel by panel before any partition is fitted. The first half holds: all five panels select a discount below one, and on the panel where both effects are measurable forgetting is worth an order of magnitude more than refinement. The second half fails in a more informative way than we expected. Refining the pooling structure improves external accuracy on none of the five panels, and a controlled surface over leverage and group separation says why. Four panels sit where a supplied true partition would earn nothing; one, Carparts, sits where it would earn 1.7 to 2.0%, and its learned partition returns  $-0.08\%$ . The same mixture objective loses at those coordinates in simulation, so on the one panel with room what binds is learnability, and on the other four it is the absence of room. Separating those two explanations is what the diagnostic is for.

The resulting empirical position is deliberately nar-

row. Across the five panels and both protocols, EBB leads on Auto and RAF, and the official Tweedie–Gaussian-process implementation leads on the others by margins that are under one percent in four of five cases and 2.9% on M5 at fixed origin; under walk-forward updating, adaptive conformal inference leads on M5 by 4.4%. Measured as the largest shortfall to the best method on any panel, EBB (2.9% at fixed origin, 4.6% under walk-forward updating) and the specialist (3.1%, 4.5%) are alone within five percent of the best competing method everywhere, and every other method is more than 15% behind on at least one panel. The adaptive conformal layer that makes ADIDA competitive can be applied to EBB as well: it takes M5 and Auto and closes Carparts to 0.2%, but degrades Online Retail and RAF; a validation-tail rule inside the initialization window decides where to apply it and is right on four of five panels. The margins are concentrated in the upper quantiles: below the median, several methods on these panels coincide exactly with the all-zero forecast. We report this structure rather than an aggregate win because it is what the mechanism predicts, and because two widely used evaluation conventions actively obscure it.

**Contributions.** (1) **Joint adaptation.** We introduce a scalable empirical-Bayes model that applies one recency operator consistently to item- and group-level sufficient statistics, then selects the discount jointly with the pooling structure by leakage-free chronological validation. The resulting predictive law is closed form, bit-reproducible, costs milliseconds per series on a CPU, and updates online without refitting. (2) **An estimable resolution condition.** We show that discounting has two linked effects: it keeps prior leverage active but reduces the information with which group heterogeneity can be estimated. A two-sided credibility bound gives the collapse probability of the group variance component, and its resolution corollary converts that bound into a statistic computable from the fitting window. The statistic bounds how much any partition could be worth on a given panel before one is fitted, and separates a panel with no room from a panel whose room we fail to use. Connections to TSB, drift tracking, classical credibility, and finite-class selection are collected in the appendix. (3) **Mechanism-focused evidence, including a negative result.** Five real-data panels, five strict walk-forward evaluations, and a known-partition simulation separate gains from forgetting, gains from pooling refinement, and distortions caused by zero-favoring point metrics. Comparators include the official TweedieGP implementation, adaptive con-

<sup>1</sup>Empirical Bayes with Bounded memory: under the recency operator an item’s effective history stays bounded (Section 4), so evidence ebbs rather than accumulates.

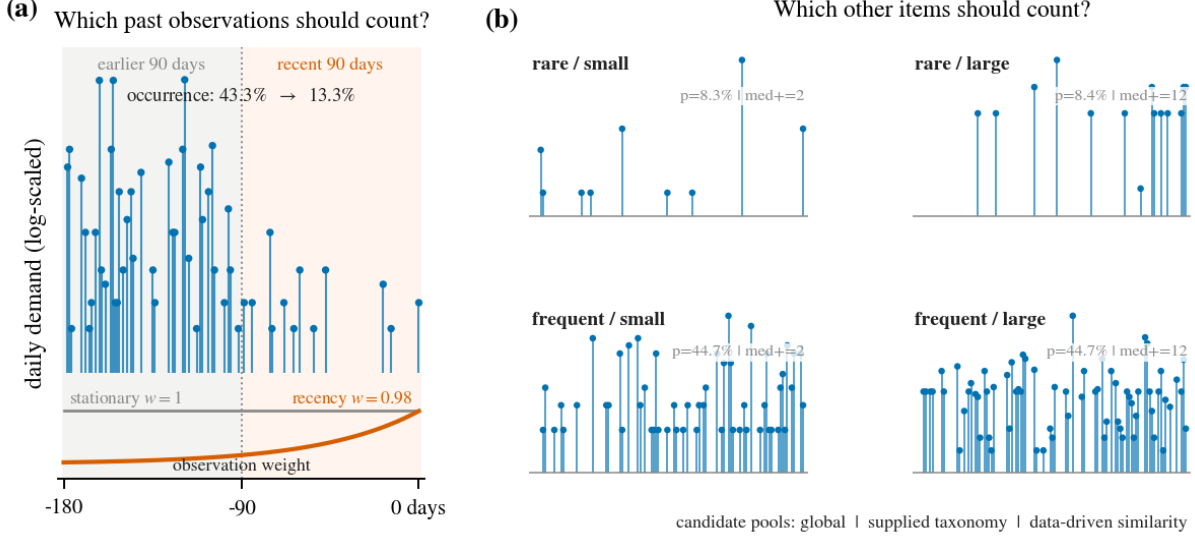


Figure 1: The two coupled choices, in real intermittent demand. **(a)** One item’s daily history: occurrence falls from 43.3% to 13.3% between the earlier and the recent ninety days, so a stationary weighting ( $w = 1$ ) and a recency weighting ( $w = 0.98$ ) summarize the same history differently—which past observations should count? **(b)** Four items from the same panel, spanning rare and frequent occurrence at small and large sizes: any pooled prior must decide which of these heterogeneous neighbors an item should borrow strength from—which other items should count? Candidate pools are a global pool, a supplied taxonomy, and data-driven similarity.

formal inference applied to the baselines and to EBB alike, a published hierarchical state-space model, and a zero-shot foundation model. Learned pooling improves external accuracy on none of the five panels; a controlled surface over leverage and group separation shows that four panels have no room and that on the fifth, Carparts, both the real objective and its simulated counterpart fail to collect it. The two obvious model-side remedies for the remaining shortfall to the specialist, heavier-tailed size laws and item-specific forgetting rates, do not close it either. We also document two evaluation failures that the intermittent-demand literature currently inherits—absolute-error metrics reward the all-zero forecast, and static split-conformal wrappers understate what conformal prediction can do here by about 20%.

## 2 Related Work

**Intermittent-demand forecasting.** Croston’s decomposition [5], its SBA correction [23], and TSB [25] estimate occurrence and positive size locally; TSB additionally lets the occurrence estimate decay under obsolescence. ADIDA [17] and IMAPA [12] instead reduce intermittency through temporal aggregation, while likelihood-based alternatives include inventory count models [21], iETS [22], Tweedie

Gaussian processes [6], and deep renewal processes [27]. Conformal wrappers turn point forecasters into predictive intervals, and adaptive conformal inference updates miscoverage levels under distribution shift [7]; we evaluate both the static and the adaptive form, because the difference between them is larger than most method differences reported on these panels. These approaches provide important forms of temporal adaptation, but do not jointly learn how recency-weighted evidence should be shared across items. Hierarchical models do share evidence: Chapados [2] and Seeger et al. [20] pool related count series at scale, and Pitkin et al. [18] develops a hierarchical hurdle model for retail. Their pooling structure or temporal evolution is generally specified in advance. EBB instead crosses recency and pooling choices within one occurrence–size empirical-Bayes model, and asks when that crossing can pay.

**Learning cross-series structure.** The continuum between local and global forecasting is formalized by Montero-Manso and Hyndman [16]. More recent work learns richer forms of cross-series specialization. For hierarchical forecasting, Cini et al. [4] learns a graph-pooled aggregation hierarchy aligned with the forecasting objective. Moirai-MoE [14] replaces human-defined frequency groups with token-level sparse-expert routing in a time-series foundation

model. These approaches motivate data-adaptive structure, but address different objects: aggregation coherence or latent expert specialization. Our groups are explicit exchangeability pools for a shared prior. That restriction is what makes their statistical resolution analyzable: the quantity governing whether a pool can be estimated is the same credibility weight that governs whether it can matter, so a single diagnostic covers both. It also permits a direct comparison among global, fixed-rule demand-pattern, and learned mixture pools under one predictive law.

**Discounting, credibility, and selection.** Exponential discounting is classical in Bayesian dynamic models [26], and power priors down-weight historical likelihoods [9]. Likewise,  $\lambda_i = n_i/(n_i + \phi_g)$  is the classical actuarial credibility form [1]; Proposition 6 is therefore a restatement, not a claimed contribution. The distinction in EBB is that one recency operator updates both item statistics and the empirical-Bayes statistics defining the shared prior, and its strength is selected jointly with the pool. Proposition 2 then identifies the resulting two-sided resolution limit: forgetting preserves prior leverage but can erase the information needed to distinguish fine pools. Chronological validation selects among the finite candidate family. Because zero-heavy point losses can favor degenerate forecasts, selection uses pinball loss, a proper scoring rule [8], with the high quantiles relevant to inventory decisions [11].

### 3 Ebb: Recency-Weighted Hierarchical EB

EBB combines two candidate choices inside one occurrence-size model: a recency discount  $w$  determines which observations remain informative, and a partition  $g$  determines which items share an empirical-Bayes prior. The key construction is that the *same* discounted sufficient statistics determine both the item posterior and the group prior. We first define that representation, then the candidate partitions and their leakage-safe joint selection.

#### 3.1 Occurrence-size model

For item  $i$ , let  $z_{i,t} = \mathbf{1}\{y_{i,t} > 0\}$  and define  $\ell_{i,t} = \log y_{i,t}$  when  $z_{i,t} = 1$ . Conditional on membership

$g_i = g(i)$ , the hurdle model is

$$\begin{aligned} \pi_i &\sim \text{Beta}(\alpha_{g_i}, \beta_{g_i}), & z_{i,t} \mid \pi_i &\sim \text{Bern}(\pi_i), \\ \mu_i &\sim \mathcal{N}(\mu_{0,g_i}, \tau_{g_i}^2), \\ \sigma_i^2 &\sim \text{ScaleInv-}\chi^2(\nu, s_{0,g_i}^2), \\ \ell_{i,t} \mid z_{i,t}=1, \mu_i, \sigma_i^2 &\sim \mathcal{N}(\mu_i, \sigma_i^2). \end{aligned} \quad (1)$$

Here  $s_{0,g}^2$  is parameterized so that the prior mean of  $\sigma_i^2$  is the group estimate  $\sigma_g^2$ ;  $\nu$  is fixed rather than selected. Group hyperparameters are estimated by empirical Bayes: Beta-Binomial marginal likelihood for occurrence, and an intercept-only random-effects fit plus conjugate variance shrinkage for positive log-size. Appendix D gives the numerical objectives and fallback rules.

Without forgetting, all inference depends on the four statistics

$$S_i(1) = (n_i, c_i, u_i, v_i) = \sum_{t=1}^{T_i} (1, z_{i,t}, z_{i,t}\ell_{i,t}, z_{i,t}\ell_{i,t}^2),$$

where the last two terms are zero when  $z_{i,t} = 0$ . The occurrence posterior mean and the positive-size posterior location take parallel credibility forms,

$$\begin{aligned} \hat{\pi}_i &= \lambda_i^{(o)} \bar{p}_i + (1 - \lambda_i^{(o)}) \mu_{\pi,g_i}, & \lambda_i^{(o)} &= \frac{n_i}{n_i + \phi_{g_i}}, \\ \hat{\mu}_i &= \lambda_i^{(+)} \bar{\ell}_i + (1 - \lambda_i^{(+)}) \mu_{0,g_i}, & \lambda_i^{(+)} &= \frac{n_i^+}{n_i^+ + \kappa_{i,g_i}}, \end{aligned} \quad (2)$$

with  $\phi_g = \alpha_g + \beta_g$  and  $\kappa_{i,g} = \hat{\sigma}_i^2 / \tau_g^2$ . Thus a partition matters only through the prior target and strength supplied to its members.

Two variance ratios appear in what follows. Estimation uses the item-level  $\kappa_{i,g} = \hat{\sigma}_i^2 / \tau_g^2$  in (2); the analysis in Section 4.2 works with the group-level surrogate  $\kappa_g = \sigma_g^2 / \tau_g^2$ , which replaces each item's process variance by its group estimate. The two coincide when the group is homogeneous in  $\sigma_i^2$ , and the surrogate is what the collapse event of Proposition 2 is stated in terms of.

The plug-in marginal predictive distribution is zero-inflated log-normal:

$$\begin{aligned} \Pr(Y_i = 0 \mid \mathcal{D}) &= 1 - \hat{\pi}_i, \\ \log Y_i \mid (Y_i > 0, \mathcal{D}) &\sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2 + V_i), \end{aligned} \quad (3)$$

where  $V_i$  is posterior uncertainty in  $\mu_i$ . Its quantiles are analytic. For point evaluation we retain the conditional plug-in mean  $\hat{y}_i = \hat{\pi}_i \exp(\hat{\mu}_i + \hat{\sigma}_i^2/2)$ ; it is distinct from the mean of (3), which also integrates  $V_i$ .

### 3.2 Shared recency weighting

For  $w \in (0, 1]$ , replace  $S_i(1)$  by

$$S_i(w) = \sum_{t=1}^{T_i} w^{T_i-t} (1, z_{i,t}, z_{i,t} \ell_{i,t}, z_{i,t} \ell_{i,t}^2). \quad (4)$$

These fractional counts are valid weighted-likelihood statistics; the log-Gamma form of the occurrence marginal and the Normal size objective extend directly to them. Crucially,  $S_i(w)$  is used twice:

$$\begin{aligned} \hat{\eta}_g(w) &= \arg \max_{\eta_g} \sum_{i: g_i=g} \log m\{S_i(w) \mid \eta_g\}, \\ q_i^{(w)}(\theta_i) &= q_i(\theta_i \mid S_i(w), \hat{\eta}_{g_i}(w)), \end{aligned} \quad (5)$$

where  $\eta_g$  collects the group hyperparameters and  $q_i$  is the item posterior. The group prior therefore adapts to recent cross-sectional evidence rather than remaining a stationary prior applied to discounted item data. Setting  $w = 1$  recovers the stationary estimator exactly.

For occurrence, (2) becomes

$$\begin{aligned} \hat{\pi}_i(w) &= \lambda_i(w) \bar{p}_i(w) + \{1 - \lambda_i(w)\} \mu_{\pi, g_i}(w), \\ \lambda_i(w) &= \frac{n_i(w)}{n_i(w) + \phi_{g_i}(w)}. \end{aligned} \quad (6)$$

The analogous positive-size update uses discounted positive mass  $n_i^+(w)$  and the group variance ratio  $\kappa_{i,g}(w)$ .

### 3.3 Candidate pooling structures

Let  $\mathcal{R}$  contain three ways to construct  $g(\cdot)$  from the fitting data. Table 1 orders them from coarse to fine: one prior for the panel, one prior per fixed demand-pattern class, and one prior per learned component. The ladder is deliberate. The question this paper asks is not which rung is best in general—that depends on the panel—but whether the discounted data support moving up it at all. Section 4.2 gives the condition under which they do, and Section 5.2 reports what happens when they do not.

For the learned candidate, component  $k$  carries  $\eta_k = (\alpha_k, \beta_k, \mu_{0,k}, \tau_k^2, \sigma_k^2)$ . Its item marginal is the product of Beta–Binomial occurrence evidence and Normal random-effects size evidence. EM has an analytic responsibility update; its M-step maximizes responsibility-weighted EB objectives. A deterministic quantile initialization makes repeated fits seed-free. Mixture labels are built once from the fitting head and shared across the discount grid, with  $K \in \{1, \dots, 6\}$  selected by BIC at that fit. Re-learning the partition separately at every discount is

Table 1: Pooling candidates used by EBB, ordered coarse to fine. Every data-dependent label is computed from the selector’s fitting head, never its validation tail.

| Candidate | Shared-prior groups                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Global    | All items share one prior.                                                                                                                |
| Taxonomy  | Four ADI/CV <sup>2</sup> demand-pattern classes (smooth, intermittent, erratic, lumpy) using the classical thresholds 1.32 and 0.49 [24]. |
| Learned   | Hard assignments from a mixture of occurrence-size prior components; $K \in \{1, \dots, 6\}$ is selected by BIC.                          |

a natural variant—changing  $w$  changes both the item evidence and the components it can support—but we do not evaluate it here. The selected responsibilities are converted to hard labels so all three candidates share the same downstream forecaster.

### 3.4 Leakage-safe joint selection

Each training series is split chronologically into a fitting head  $\mathcal{H}_i$  (first 80%) and validation tail  $\mathcal{V}_i$  (last 20%). For every  $(r, w) \in \mathcal{R} \times \mathcal{W}$ , with

$$\mathcal{W} = \{1, .999, .997, .995, .99, .98, .95, .90\},$$

the partition builder for  $r$  sees only  $\mathcal{H} = \cup_i \mathcal{H}_i$ . The resulting labels, discounted EB prior, and item posteriors are all fit on the same head. The candidate is scored by

$$\begin{aligned} \hat{R}(r, w) &= \frac{1}{N_V} \sum_{i \in \mathcal{I}_V} \frac{1}{|\mathcal{V}_i| |\mathcal{Q}|} \\ &\times \sum_{t \in \mathcal{V}_i} \sum_{q \in \mathcal{Q}} \frac{\rho_q(y_{i,t}, \hat{Q}_{i,t}^{(r,w)}(q))}{a_i}, \end{aligned} \quad (7)$$

where  $\mathcal{Q} = \{.5, .75, .9\}$ ,  $\rho_q$  is pinball loss, and  $a_i$  is mean absolute demand on the fitting head (with a panel-level fallback for an all-zero head);  $N_V = |\mathcal{I}_V|$  is the number of scored items. Times and quantiles are averaged within item before items receive uniform weight. This matches the panel-level goal and prevents long series or high-volume items from dominating selection.

The selected pair is  $(\hat{r}, \hat{w}) = \arg \min_{r,w} \hat{R}(r, w)$ . Its partition is then *relearned* on the complete training window at  $\hat{w}$ , and the EB model is refit once before external evaluation. Thus the validation tail chooses a construction rule and discount, not a permanent set of labels; external targets are never used. There are 24 outer forecast candidates.

**Resolution diagnostics.** Fitting each candidate already produces everything the resolution analysis of Section 4.2 needs, so the selector records it rather than recomputing it later. For every  $(r, w)$  and every resulting group  $g$  we store, from the fitting head only, the median occurrence and positive-size credibility weights  $\lambda_i^{(o)}$  and  $\lambda_i^{(+)}$ , the share of items with  $\lambda_i^{(+)} < 0.1$ , the group sizes  $m_g$ , and the resolution statistic

$$\hat{z}_g = \frac{S_g^2 - \bar{s}_g}{\sqrt{2/(m_g - 1) S_g^2}},$$

whose nonpositive values coincide with the collapse event  $\hat{\tau}_g^2 = 0$  of Proposition 2. These quantities cost nothing beyond the fit that produced them, they are computed before any external observation is seen, and they answer a question the validation score cannot: not which candidate scores best on the tail, but whether the discounted data contain enough information to estimate the candidate’s group structure at all. Section 5.2 uses them to separate the two. The full selection surface and the accompanying diagnostics are released with every run (`remix_selection_diagnostics.csv`).

Appendix D gives split edge cases and pseudocode-level implementation details.

### 3.5 Forecasts and computation

Selection uses the analytic plug-in quantiles in (3). The reported probabilistic forecasts also average quantile estimates across  $B = 20$  cross-item bootstrap refits of the group hyperparameters. We treat this explicitly as a quantile ensemble; the average of component quantiles is not asserted to be the quantile of a single posterior mixture.

For each  $w$ , one pass computes  $S_i(w)$  in  $O(\sum_i T_i)$  time. Each EB fit requires only low-dimensional deterministic optimization per group; mixture EM is  $O(NK)$  per iteration. Prediction is  $O(N)$  for points and  $O(NQ)$  for  $Q$  quantiles.

Two properties of this cost profile matter for the comparisons that follow. First, the whole pipeline is CPU-only and closed form apart from the mixture search: fitting and predicting a panel costs on the order of milliseconds per series, so the full 24-candidate joint selection completes in minutes on a laptop-class machine and no GPU, tuning budget, or convergence diagnostic is required. Second, estimation is exact rather than approximate at every stage, so repeated end-to-end runs are bit-identical in the released test suite. Neither property is an accuracy claim; both are the terms on which the accuracy results in Section 5 should be read, since several of the strongest

comparators there cost two to three orders of magnitude more per series and none is reproducible to the bit. Measured wall-clock figures for every method are in Appendix D.

## 4 Resolution under Forgetting

The shared operator in (5) creates two opposing effects. A bounded item history prevents the group prior from vanishing, but the same bound limits the information used to estimate fine group structure. This section formalizes that interaction. Supporting results connecting the model to TSB, drift tracking, finite-class selection, and classical credibility are collected in Appendix B.

### 4.1 Persistent prior leverage

Fix an item and suppress its subscripts. Because  $n(w) = \sum_{t=1}^T w^{T-t} \leq (1-w)^{-1}$  for  $w < 1$ , the occurrence credibility weight cannot converge to one.

**Proposition 1** (Forgetting keeps pooling relevant). For every horizon  $T$ ,  $\lambda(w) \leq \{1 + (1-w)\phi\}^{-1} < 1$  when  $w < 1$ , whereas  $\lambda(1) \rightarrow 1$  as  $T \rightarrow \infty$ . Hence the pooled occurrence prior retains positive weight indefinitely under forgetting.

The statement is about leverage, not benefit: it does not guarantee that a particular prior is a good shrinkage target. It says that the choice of target can continue to affect forecasts even for old series.

### 4.2 Variance-component resolution

The converse problem appears in the positive-size block. Let item  $i$  in group  $g$  have discounted positive mass  $n_i^+$ , sampling variance  $s_i = \sigma_g^2/n_i^+$  for its log-size mean, and credibility  $\lambda_i^{(+)} = n_i^+/(n_i^+ + \kappa_g)$  with  $\kappa_g = \sigma_g^2/\tau_g^2$ . The group variance is estimated by the truncated moment/REML form  $\hat{\tau}_g^2 = \max(0, S_g^2 - \bar{s}_g)$ , where  $S_g^2$  is the sample variance of the  $m_g$  item means and  $\bar{s}_g = m_g^{-1} \sum_{i \in g} s_i$ .

**Proposition 2** (Two-sided effect of forgetting). (i) For  $w < 1$ ,  $\lambda_i^{(+)} \leq \{1 + (1-w)\kappa_g\}^{-1} < 1$ . (ii) For independent item means with possibly unequal  $s_i$ ,

$$\mathbb{E}[S_g^2] = \tau_g^2 + \bar{s}_g$$

holds exactly. Under the Gaussian moment approximation,

$$\Pr(\hat{\tau}_g^2 = 0) \simeq \Phi\left(-\sqrt{\frac{m_g - 1}{2}} \rho_g\right), \quad \rho_g = \frac{\tau_g^2}{\tau_g^2 + \bar{s}_g}. \quad (8)$$

Since  $n_i^+ \leq (1-w)^{-1}$ ,  $\bar{s}_g \geq (1-w)\sigma_g^2$  and therefore

$$\rho_g \leq \frac{\tau_g^2}{\tau_g^2 + (1-w)\sigma_g^2}.$$

On the collapse event the plug-in estimate has  $\hat{\tau}_g^2 = 0$  and  $\lambda_i^{(+)} = 0$ : the item mean is replaced by its estimated group mean.

The population ratio  $\rho_g$  is not observed. For diagnostics we use

$$\hat{z}_g = \frac{S_g^2 - \bar{s}_g}{\sqrt{2/(m_g - 1)} S_g^2},$$

whose nonpositive values coincide with the plug-in collapse event.

**Corollary 1** (Resolution condition). For this estimator and window, stable estimation of a group variance requires, up to a fixed probability threshold,

$$\rho_g \sqrt{m_g - 1} \gtrsim 1, \quad \text{or equivalently} \quad \tau_g^2 \gtrsim \frac{\bar{s}_g}{\sqrt{m_g - 1}}$$

with  $\bar{s}_g \geq (1-w)\sigma_g^2$ .

For similarity-based refinements, finer groups typically reduce both  $m_g$  and the within-group spread  $\tau_g^2$ , while stronger forgetting raises the sampling-variance floor. All three changes move the estimator toward collapse. This is an estimator- and window-specific resolution condition, not an information-theoretic lower bound, and it does not say which candidate wins below the threshold.

Figure 2 summarizes why pooling can become more consequential at exactly the point where a fine partition becomes harder to estimate. Section 5.5 tests both sides with known group labels. Proofs and the accuracy limits of the approximation are given in Appendix C and Section 6.

## 5 Experiments

### 5.1 Experimental design

**Datasets.** We evaluate on two daily panels—**Online Retail** [3] (3,649 series) and a seed-controlled 5,000-series **M5** sample [15]—and the monthly intermittent suite of Damato et al. [6]: **Auto** (3,000 series), **Carparts** (2,509), and **RAF** (5,000). Together they contain more than 19,000 series and span long daily histories and short, highly sparse monthly panels. Dataset construction and holdout lengths are in Appendix E.16. Both the probabilistic and the point

M5 protocols use the first two thirds of each series for initialization.

**Comparators.** The classical set comprises Croston, SBA, TSB, ADIDA, and IMA; stronger statistical comparators are AutoARIMA, AutoTheta, a per-series autoregressive Tweedie GLM, iETS, and the official TweedieGP implementation at its released defaults on all five panels. For probabilistic evaluation we use native distributions where available and chronological split-conformal wrappers (CP-\*) for the point forecasters. The walk-forward studies additionally include adaptive conformal inference (ACI) applied to ADIDA and to EBB on every panel and to TSB on Online Retail; ACI-EBB reads EBB’s analytic predictive quantile at the adaptive level, with the same  $\gamma$  and update rule as ACI-ADIDA; EBB (rule) applies that layer only where a validation-tail rule inside the initialization window selects it (Section 5.3). A 25-candidate tuned-TSB control receives the same internal chronological selection budget as EBB. DeepState and Chronos-Bolt comparisons are confined to Appendix E.13 because they use subset or proxy protocols. Implementation details and complete settings are in Appendix D.

**Protocols and metrics.** The primary fixed-origin experiment fits once on the initialization window and scores the full evaluation span. A strict walk-forward experiment then reveals and updates one block at a time: seven-day blocks on Online Retail and M5, one-month blocks on Auto, Carparts, and RAF. Classical, conformal, and TweedieGP methods refit each block, except that on Online Retail AutoARIMA, AutoTheta, iETS, and TweedieGP refit every four blocks and on M5 only the conformal and adaptive-conformal wrappers are run (Appendix E.7 gives the cost of the rest); EBB updates discounted sufficient statistics each block. The primary metric is scaled pinball loss (SPL; lower is better), averaged over  $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ ; we separately report  $q_{90}$  and the far tail because inventory decisions are asymmetric. All quantiles are zero-truncated and monotonized, and EBB is reported without post-hoc calibration. Every model in Table 2 is scored by one implementation from its stored quantile predictions, so no cell mixes scoring versions. Fixed-origin, walk-forward, and subset foundation model results are kept separate because they answer different questions. MAE and RMSSE are secondary diagnostics rather than the headline criteria because zero forecasts can minimize absolute error on intermittent panels. Full point, interval, significance, and efficiency results are in Appendix E.

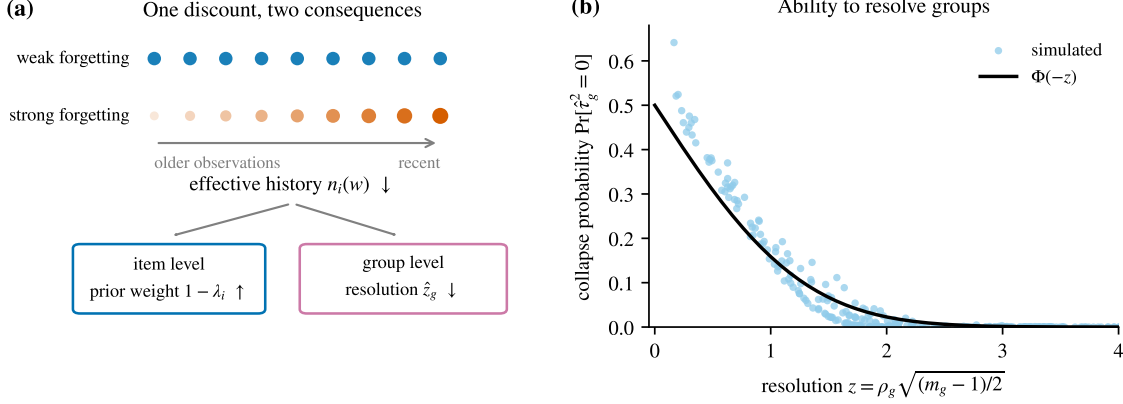


Figure 2: **One discount creates two opposing effects.** (a) The same recency operator reduces item effective history, which increases prior leverage, and group effective information, which reduces structural resolution. (b) When the partition must be estimated, the probability that the group variance collapses rises as the resolution statistic falls; the curve is the approximation in (8).

## 5.2 Probabilistic forecasting across panels

Table 2 sets the empirical position, and it is a narrow one. EBB attains the lowest mean SPL on Auto and RAF, by 1.6% and 3.0% over the official TweedieGP implementation, both significant (Table 21); TweedieGP attains it on Online Retail, Carparts, and M5, by 0.05%, 0.5%, and 2.9%, of which only the M5 margin is significant. Against every other comparator EBB is best on all five panels, by 10.1% over iETS on Online Retail and 1.2% over CP-Croston on M5. Across three independent 5,000-series M5 samples the selected configuration is identical—(mixture,  $w=0.95$ ,  $K=6$ )—and EBB has mean SPL  $1.7501 \pm 0.084$  against  $1.7547 \pm 0.065$  for CP-Croston, its closest comparator: ahead on two samples, behind by 1.0% on the third, and ahead on average by 0.3%, well within one standard deviation. We report the spread rather than an aggregate win because it is the quantity Section 5.6 explains. The explanation is not that these panels leave no room for a hierarchical construction to work; on three of them the credibility weights say otherwise. It is that the room which exists is not converted by a learned partition, which is why the achievable margin is set by the shared predictive law rather than by the pooling structure.

The degenerate row exposes what the central quantiles cannot see. On RAF, the all-zero forecast is marginally better through q90 because the true quantiles of most series are zero. At  $q \in \{0.95, 0.975, 0.99\}$  it falls behind every intensity-bearing method, and intensity-bearing models become distinguishable

from one another (Appendix E.14). TweedieGP is strongest throughout the Carparts far tail and remains competitive on RAF. The probabilistic claim is therefore not that one distribution wins every quantile, but that EBB is the most consistent aggregate performer across panels and remains informative in the region where a zero forecast ceases to be useful.

Point forecasts tell a deliberately weaker story. EBB is best on six of ten MAE/RMSSE cells and top two on seven, but the zero forecast has the lowest MAE on three panels and classical methods lead several remaining cells. We report the complete point table and horizon decomposition in Appendix E.1: they establish competitiveness and rule out a long-horizon averaging artifact, but do not support uniform point-forecast superiority.

## 5.3 Strict walk-forward forecasting

Table 3 is the deployment-facing test, and it reproduces the fixed-origin picture for the uncalibrated methods. EBB leads on Auto and RAF, by 0.5% and 4.3% over TweedieGP; TweedieGP leads on Online Retail and Carparts by 0.2% and 0.8%. Per-series paired tests (Table 22) cannot separate the two on Online Retail, Carparts, or Auto, and favor EBB on RAF. Over adaptive conformal inference on ADIDA, EBB gains 6.2% on Online Retail and 8.0% on Carparts, beats every static conformal wrapper on all five panels, and loses M5 by 4.4%.

The M5 loss is a calibration effect, and the same layer is available to EBB. ACI-EBB reads EBB’s predictive quantile at the adaptive level instead of the nominal one. On M5 it lowers mean SPL from 1.4851

Table 2: **Probabilistic forecasting is the primary comparison.** Scaled pinball loss (SPL; lower is better) is reported as the mean over  $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$  and at q90. Bold and underline mark the best and second-best non-degenerate methods. M5 reports seed 42 for direct comparability; three-seed robustness is summarized in the text. EBB uses its raw predictive law, with no post-hoc calibration. TweedieGP is the official implementation of Damato et al. at its released defaults on all five panels; Appendix E.7 reports measured cost. The all-zero row is a metric diagnostic and is excluded from ranking.

| Model             | Online Retail |               | M5            |               | Auto          |               | Carparts      |               | RAF           |               |
|-------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                   | Mean          | q90           | Mean          | q90           | Mean          | q90           | Mean          | q90           | Mean          | q90           |
| AutoARIMA         | 1.0069        | 1.6297        | 1.7919        | 2.4148        | 0.3189        | 0.2659        | 0.4084        | 0.4897        | 0.4020        | 0.5910        |
| AutoTheta         | 1.0055        | 1.6056        | 1.7954        | 2.3123        | 0.3403        | 0.2760        | 0.3925        | <u>0.4697</u> | 0.4104        | 0.5956        |
| CP-Croston        | 1.5502        | 2.7686        | 1.6902        | <b>2.1551</b> | 0.3260        | 0.3105        | 0.4012        | 0.5433        | 0.3322        | 0.4927        |
| CP-SBA            | 1.5492        | 2.8601        | 1.6939        | 2.1736        | 0.3256        | 0.3153        | 0.3982        | 0.5418        | 0.3290        | 0.4920        |
| CP-TSB            | 1.1744        | 2.6746        | 1.8794        | 2.3148        | 0.3459        | 0.3150        | 0.3783        | 0.4842        | 0.3814        | 0.4986        |
| CP-ADIDA          | 1.1239        | 2.5409        | 1.7150        | 2.1949        | 0.3275        | 0.3082        | 0.3571        | 0.4982        | 0.3097        | 0.4905        |
| CP-IMAPA          | 1.1280        | 2.5506        | 1.7190        | 2.1904        | 0.3279        | 0.3084        | 0.3561        | 0.4908        | 0.3116        | 0.4907        |
| Tweedie-GLM       | 1.1477        | 1.7458        | 4.0693        | 5.2620        | 0.3221        | 0.2783        | 0.4776        | 0.6897        | 0.3121        | 0.5312        |
| iETS              | 0.9846        | 1.7544        | 2.0282        | 2.9772        | 0.3448        | 0.3252        | 0.4561        | 0.5646        | 0.3494        | <u>0.4900</u> |
| TweedieGP         | <b>0.8845</b> | <b>1.5218</b> | <b>1.6228</b> | <u>2.1723</u> | <u>0.2985</u> | <b>0.2598</b> | <b>0.3214</b> | <b>0.4600</b> | <u>0.2763</u> | 0.5219        |
| Zero (diagnostic) | 0.9161        | 1.6490        | 1.9590        | 3.5262        | 0.5612        | 1.0102        | 0.3736        | 0.6725        | 0.2669        | 0.4805        |
| EBB (ours)        | <u>0.8849</u> | <u>1.5268</u> | <u>1.6707</u> | 2.4758        | <b>0.2937</b> | <u>0.2599</u> | <u>0.3229</u> | 0.4817        | <b>0.2680</b> | <b>0.4856</b> |

to 1.3785, 2.9% below ACI-ADIDA and significant; it also leads Auto (0.2901) and brings Carparts to within 0.2% of TweedieGP. On Online Retail and RAF the same layer hurts, by 6.3% and 2.3%, because the per-series coverage signal is too noisy on those panels to improve an already calibrated forecaster. Whether to apply it is therefore a selection problem, and the initialization window already answers it: on the same 80/20 chronological split used to select the discount, EBB is walked forward over the validation tail with and without the layer, and the layer is applied iff its tail pinball is lower (Appendix 23). The rule applies the layer on M5 and Auto and withholds it elsewhere, matching the ex-post better variant on 4 of five panels (Carparts is the exception, where the two are within 0.5% and statistically undecided), and the tail gains it sees track the external ones (+7.8% versus +7.2% on M5, −7.9% versus −6.3% on Online Retail). The resulting row, EBB (rule), is within 0.8% (Carparts) of the best competing method on every panel under walk-forward updating. Two further facts survive the mixed result: under equal calibration EBB beats the ADIDA base on all five panels (undecided only on Online Retail), and the adaptive layer’s benefit is a property of the panel, not of the forecaster it wraps.

Table 4 and Figure 3 summarize both protocols by the largest shortfall to the per-panel best. Against competing methods, EBB and TweedieGP are the only ones within five percent everywhere (EBB at

most 3.0% at fixed origin and 4.6% under walk-forward updating; TweedieGP 3.1% and 4.5%), and every other method is at least 15% behind on some panel. Counting our own wrapper as a competitor, uncalibrated EBB trails ACI-EBB by 7.7% on M5, and EBB (rule) closes that to 0.8%. The two leaders differ in cost. TweedieGP refits per-series variational GPs at every update, so its walk-forward runs took between 36 minutes (Auto) and 2.4 hours (RAF) with twelve workers and project to about 51 hours on M5 at the Online Retail cadence; EBB updates sufficient statistics in closed form and completes each of the five walk-forward runs in under 40 seconds on one core, and ACI-EBB adds a few seconds (Table 27).

Three features of this table deserve more attention than the margins. First, *the choice of conformal construction matters more here than the choice of forecaster*: making the wrapper adaptive [7] improves ADIDA’s mean SPL from 1.1576 to 0.9108, a 21% gain that is more than three times the remaining gap to EBB. Static split-conformal wrappers are the standard probabilistic baseline in this literature; on this evidence they understate what conformal prediction can do on intermittent panels, and comparisons drawn against them—including earlier versions of our own—are correspondingly generous. Second, *coverage and sharpness separate*. ACI-ADIDA attains higher positive-demand coverage than EBB on Online Retail (0.646 versus 0.621) with intervals almost twice as wide (19.67 versus 10.31; Table 11); neither

Table 3: **Strict probabilistic walk-forward evaluation on all five panels.** Online Retail and M5 use seven-day blocks; Auto, Carparts, and RAF use monthly blocks. Mean and q90 are scaled pinball losses; all five quantiles, coverage, and interval width are reported in Appendix E.3. Classical wrappers and ACI refit each block. AutoARIMA, AutoTheta, iETS, and TweedieGP refit every four blocks on Online Retail and every block on the monthly panels; on M5 these four were not run under this protocol on cost grounds (dashes). EBB updates sufficient statistics each block; ACI-EBB applies the ACI-ADIDA calibration layer to EBB’s predictive quantiles, and EBB (rule) applies it only where the validation-tail rule selects it, so it equals one of the two rows above it on each panel and is not ranked. Bold and underline rank non-degenerate methods by SPL; on RAF the all-zero forecast is marginally lower than every method through q90. On M5, the longest daily panel, the adaptive conformal layer decides the ranking: ACI-EBB leads, ACI-ADIDA is second, and uncalibrated EBB is third.

| Model             | Online Retail |               | Carparts      |               | Auto          |               | RAF           |               | M5            |               |
|-------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                   | Mean          | q90           | Mean          | q90           | Mean          | q90           | Mean          | q90           | Mean          | q90           |
| AutoARIMA         | 1.0627        | 1.5225        | 0.4195        | 0.4839        | 0.3139        | 0.2632        | 0.4083        | 0.5920        | —             | —             |
| AutoTheta         | 1.0586        | 1.4835        | 0.4056        | 0.4743        | 0.3224        | 0.2619        | 0.4308        | 0.6233        | —             | —             |
| iETS              | 1.1092        | 1.9404        | 0.4362        | 0.5154        | 0.3286        | 0.3009        | 0.3537        | 0.4902        | —             | —             |
| CP-TSB            | 1.3733        | 2.8422        | 0.3836        | 0.4758        | 0.3297        | 0.3009        | 0.3842        | 0.4941        | 1.6616        | 2.0422        |
| CP-ADIDA          | 1.1576        | 2.5143        | 0.3509        | 0.4880        | 0.3175        | 0.2949        | 0.3095        | <u>0.4894</u> | 1.5537        | 1.9629        |
| CP-IMAPA          | 1.1773        | 2.5224        | 0.3492        | 0.4789        | 0.3176        | 0.2948        | 0.3127        | 0.4903        | 1.5457        | 1.9490        |
| ACI-ADIDA         | 0.9108        | 1.6681        | 0.3417        | 0.4767        | 0.3147        | 0.2900        | 0.3085        | 0.5598        | <u>1.4192</u> | <u>1.8064</u> |
| TweedieGP         | <b>0.8526</b> | <b>1.3791</b> | <b>0.3121</b> | <b>0.4326</b> | 0.2924        | <b>0.2539</b> | 0.2801        | 0.5261        | —             | —             |
| Zero (diagnostic) | 0.9161        | 1.6490        | 0.3736        | 0.6725        | 0.5612        | 1.0102        | 0.2669        | 0.4805        | 1.9590        | 3.5262        |
| EBB               | <u>0.8540</u> | <u>1.4002</u> | 0.3145        | 0.4578        | <u>0.2910</u> | 0.2563        | <b>0.2681</b> | <b>0.4864</b> | 1.4851        | 1.8797        |
| ACI-EBB           | 0.9082        | 1.5623        | <u>0.3128</u> | <u>0.4521</u> | <b>0.2901</b> | <u>0.2560</u> | <u>0.2743</u> | 0.5175        | <b>1.3785</b> | <b>1.6438</b> |
| EBB (rule)        | 0.8540        | 1.4002        | 0.3145        | 0.4578        | 0.2901        | 0.2560        | 0.2681        | 0.4864        | 1.3785        | 1.6438        |

method dominates, and the SPL ordering reflects sharpness rather than calibration. Third, *the central quantiles carry little information on the sparse panel*: on Carparts the q10 and q25 entries of EBB equal the all-zero forecast exactly and q50 sits marginally below it, so essentially the entire mean-SPL difference is generated above the median. We return to this in Appendix E.14, where the far-tail levels are the only ones that separate intensity-bearing models on the sparsest panels.

The configuration is selected once at the origin and then updated online; repeated online re-selection is outside the present experiment.

## 5.4 Short histories

Corollary 1 predicts where a pooled prior pays: when an item’s own evidence is thin, its credibility weight is low and the shared prior carries the forecast. Truncating the initialization window to its last  $L$  observations per series, with the evaluation span unchanged, moves every panel along that axis. Figure 4 plots EBB’s mean-SPL advantage over TweedieGP, the strongest per-series comparator, against the retained history; Table 17 gives the numbers together with the median credibility weights and the static conformal wrapper. The advantage grows as history shrinks on Auto and RAF, turns positive on Carparts only at

the shortest window, stays within a third of a percent on Online Retail, and does not appear on M5, where both median credibility weights stay above 0.8 even with 28 days of history and the per-series GP leads at every length but one, by a margin that grows with the history it is given. In words: Auto: +1.6% at full length, +6.6% with 6 periods; RAF: +3.0% at full length, +11.3% with 6 periods; Carparts: -0.5% at full length, +2.4% with 6 periods; Online Retail: -0.0% at full length, +0.3% with 14 periods; M5: -3.0% at full length, -1.8% with 28 periods. Both methods degrade gracefully; the gap that opens at short histories is the pooled prior’s contribution, and its size tracks the leverage the diagnostic assigns to each panel rather than the panel’s frequency.

## 5.5 Controlled test of the resolution mechanism

Real data never reveal the true pooling partition, so on their own they cannot distinguish a partition that is not worth having from one we failed to find. Both appear as a small number. We therefore construct panels where the partition is correct by definition, and use them as a reference against which the learned partitions of Section 5.6 are measured: what a good partition is worth at a given level of credibility leverage, rather than what our estimator obtains there.

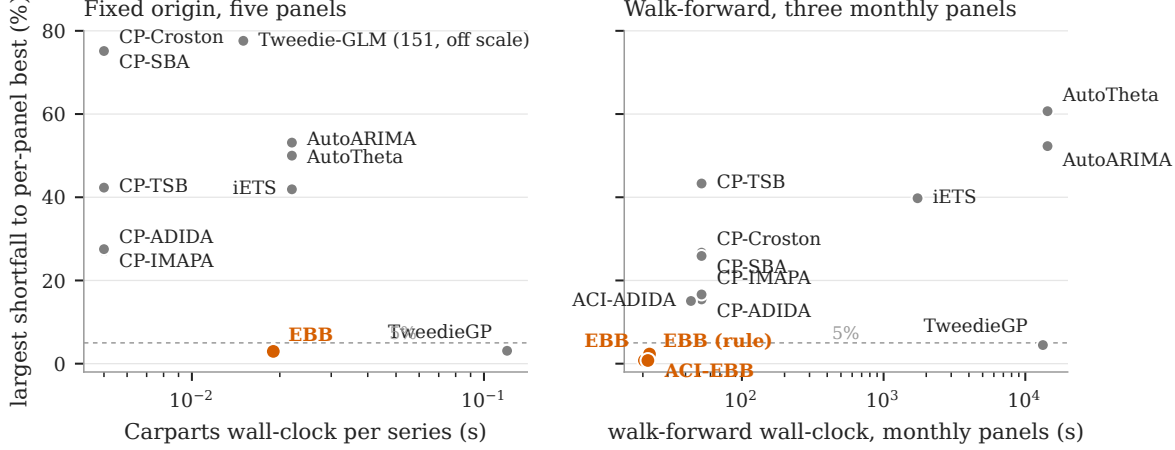


Figure 3: **Largest shortfall against cost.** Each point is one method;  $y$  is its largest mean-SPL shortfall to the per-panel best (Table 4),  $x$  is measured wall-clock on a log scale. Left: fixed origin over five panels against Carparts cost per series (Table 26). Right: walk-forward over the three monthly panels, on which every method has a measured wall-clock, against the total walk-forward wall-clock on those panels (Table 27; TweedieGP uses twelve workers, every other method one process). The dashed line marks 5%.

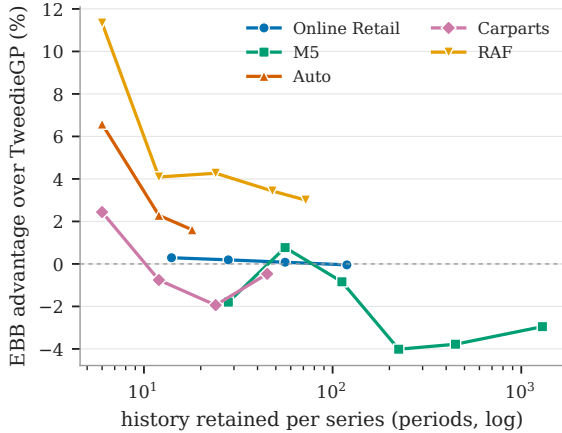


Figure 4: **Pooling pays when histories are short.** EBB’s mean-SPL advantage over TweedieGP (percent) against the history retained per series at initial-ization; the evaluation span is fixed and the audited structure and discount are unchanged. Full-length points reproduce Table 2.

We simulate from the hierarchical hurdle model of (1) with group centers arranged symmetrically around a fixed grand mean, so that increasing separation does not change the panel scale. Each cell compares a single pool against the known true partition under the same predictive representation, with nine paired replicates, and sweeps the two quantities that Section 4.2 identifies as controlling credibility:

the discount  $w$  and the series length  $T$ . The resulting grid spans the leverage range the real panels occupy, from the near-fully-shrunk regime of Carparts to the item-dominated regime of Auto and RAF.

Table 5 gives the result. The true partition improves NLL and log-size MSE in every cell, and pinball loss whenever forgetting is strong; in a separate length sweep its pinball advantage decays from 1.53% at  $T = 12$  to 0.01% at  $T = 400$ . Both patterns follow the credibility mechanism directly. Discounting or short histories reduce an item’s own weight  $\lambda_i^{(+)}$  and open room for a prior to act; long undiscounted histories drive  $\lambda_i^{(+)} \rightarrow 1$  and make every partition irrelevant. Stronger forgetting simultaneously makes group variance components harder to estimate, as Proposition 2 predicts, which is why the gains in the  $w = 0.90$  rows are bought against a noisier  $\hat{\tau}_g^2$  and do not continue to grow without limit. Extending the grid with a separation axis (Appendix E.8; group centers 0 to 3 log-units apart) shows that the return is governed by the forgetting regime as much as by leverage: with no separation in either block the return is within noise of zero; with  $w = 0.90$  a true partition earns 1–2.7% from separation 0.5 upward, with  $w = 0.95$  it exceeds 1% only at separation 0.5 with long histories ( $\lambda^{(+)} \approx 0.4$ ), and with  $w = 1$  it never does. The learned mixture, run on the same simulated panels, is negative at every moderate separation, and the two remedies already in the code base—label sample-splitting and credibility shrinkage of the group variance—do not recover the loss

Table 4: **Largest shortfall to the per-panel best.** For each method, the maximum over panels of its mean-SPL regret to the best non-degenerate method on that panel (percent; the panel attaining it in parentheses; Carp. is Carparts), and the number of panels on which the method is within 5% of the best. A superscript gives the number of panels evaluated when fewer than five. ACI-EBB and the rule row exist only under walk-forward updating; the rule row is excluded from the per-panel best. Per-panel regrets are in Table 16.

| Method      | Fixed origin |            | Walk-forward              |            |
|-------------|--------------|------------|---------------------------|------------|
|             | Max (panel)  | $\leq 5\%$ | Max (panel)               | $\leq 5\%$ |
| AutoARIMA   | 50.0 (RAF)   | 0/5        | 52.3 (RAF) <sup>4</sup>   | 0/4        |
| AutoTheta   | 53.1 (RAF)   | 0/5        | 60.7 (RAF) <sup>4</sup>   | 0/4        |
| Tweedie-GLM | 150.8 (M5)   | 0/5        | —                         | —          |
| iETS        | 41.9 (Carp.) | 0/5        | 39.8 (Carp.) <sup>4</sup> | 0/4        |
| CP-Croston  | 75.3 (OR)    | 1/5        | 60.9 (OR)                 | 0/5        |
| CP-SBA      | 75.1 (OR)    | 1/5        | 61.5 (OR)                 | 0/5        |
| CP-TSB      | 42.3 (RAF)   | 0/5        | 61.1 (OR)                 | 0/5        |
| CP-ADIDA    | 27.1 (OR)    | 0/5        | 35.8 (OR)                 | 0/5        |
| CP-IMAPA    | 27.5 (OR)    | 0/5        | 38.1 (OR)                 | 0/5        |
| ACI-ADIDA   | —            | —          | 15.1 (RAF)                | 1/5        |
| TweedieGP   | 3.1 (RAF)    | 5/5        | 4.5 (RAF) <sup>4</sup>    | 4/4        |
| EBB         | 3.0 (M5)     | 5/5        | 7.7 (M5)                  | 4/5        |
| ACI-EBB     | —            | —          | 6.5 (OR)                  | 4/5        |
| EBB (rule)  | —            | —          | 0.8 (Carp.)               | 5/5        |

(grid means  $-0.58\%$ ,  $-0.79\%$ ,  $-0.66\%$ , and  $-0.56\%$  combined).

MAE moves the other way in all six cells. A known-correct partition, on data generated from the model it describes, is reported as harmful by the metric most commonly used in this literature. We take this as the cleanest available statement of why Section 5.2 treats absolute-error criteria as diagnostics rather than as the headline.

**What these panels are for.** Plotted against median  $\lambda^{(+)}$ , the cells of Table 5 and the length sweep trace the grey curves of Figure 5: the return to *correct* structure as a function of leverage. Because the partition is supplied rather than estimated, that return is an upper envelope—no procedure that must also learn the grouping can exceed it, and any procedure that falls below it is losing something to estimation rather than to a lack of structure. This is the comparison Section 5.6 draws, and it is the reason the controlled study is necessary rather than decorative: the real panels supply the leverage coordinates, and only a known-partition experiment can supply the ordinate they should be judged against.

Two boundaries apply. The experiment isolates the value of correct structure and says nothing about

Table 5: Advantage of the *true* partition over a single pool on simulated panels (mean percentage improvement over nine paired replicates; standard error in parentheses). Positive favors the partition.  $T$  is series length.

| $T$ | $w$  | NLL         | log-size MSE | pinball      | MAE          |
|-----|------|-------------|--------------|--------------|--------------|
| 30  | 0.90 | 2.40 (0.18) | 17.15 (2.39) | 2.25 (0.61)  | -1.72 (0.73) |
| 30  | 0.95 | 1.12 (0.39) | 9.00 (2.45)  | -0.44 (0.58) | -0.17 (0.51) |
| 30  | 1.00 | 1.02 (0.08) | 8.37 (1.53)  | -0.69 (0.32) | -1.95 (0.59) |
| 120 | 0.90 | 1.65 (0.22) | 18.36 (1.56) | 1.57 (0.64)  | -3.34 (1.37) |
| 120 | 0.95 | 0.96 (0.11) | 3.65 (0.77)  | 0.23 (0.28)  | -1.32 (0.73) |
| 120 | 1.00 | 0.23 (0.04) | 1.13 (0.31)  | -0.32 (0.13) | -0.80 (0.11) |

whether the mixture recovers it—indeed Section 5.6 shows it does not, and the gap between the two is precisely what the comparison measures. And it simulates from the model that EBB fits, so the mechanism is tested under correct specification only; a misspecified size law would shift the envelope, though it is hard to see how it would raise it. Full diagnostics, including the growth of  $\text{RMSE}(\hat{\tau}_g^2)$  as  $w$  falls, are in Appendix E.8.

## 5.6 When can pooling pay?

The results so far are consistent and small. This section asks whether their size is a property of the data or of our estimator, and shows that the two can be told apart with quantities the selector already computes on the fitting head (Section 3.4).

**What gets selected, and how weakly.** All five panels select  $w < 1$ : Carparts 0.90, Online Retail and M5 0.95, Auto 0.99, and RAF 0.997. Across three independent 5,000-series M5 samples the selected configuration is identical in structure, discount, and mixture order—(mixture,  $w=0.95$ ,  $K=6$ ) in every case.

The pooling axis behaves differently, and the honest summary is that it barely resolves. On three panels the spread between the best and worst pooling structure at the selected discount is under a tenth of a percent of validation SPL: 0.05% on Online Retail, 0.08% on Carparts, 0.02% on RAF. Only two panels separate at all—M5, where the learned mixture beats the global pool by 0.58%, and Auto, where the global pool beats every refinement by 0.31%. Announcing a winner on the other three would report selector noise. What the selector reliably does is choose a discount and avoid poor structural corners; it does not discover demand taxonomies, and we do not claim it does.

Table 6: **Credibility leverage under a single pool.** Median credibility weight per panel at no forgetting and at the selected discount, for both blocks of the hurdle model:  $\lambda_i^{(o)} = n_i/(n_i + \phi_g)$  for occurrence and  $\lambda_i^{(+)} = n_i^+/(n_i^+ + \kappa_g)$  for positive size, the latter being the quantity bounded by Proposition 2. A weight near one means the item’s own history dominates its posterior, bounding how far any shared prior can move it. “< 0.1” is the share of items whose size-block weight falls below 0.1. The final column is the external MAE change of a separately selected learned partition relative to the single pool; negative favors the single pool.

| Panel         | $w^*$ | $\lambda^{(+)} \text{ (size)}$ |              |       | $\lambda^{(o)} \text{ (occ.)}$ |              |
|---------------|-------|--------------------------------|--------------|-------|--------------------------------|--------------|
|               |       | $w=1$                          | $w^*$        | < 0.1 | $w=1$                          | $w^*$        |
| Carparts      | 0.90  | 0.802                          | <b>0.224</b> | 22.5% | 0.882                          | 0.628        |
| Online Retail | 0.95  | 0.897                          | <b>0.553</b> | 14.6% | 0.969                          | 0.895        |
| M5            | 0.95  | 0.996                          | 0.841        | 12.9% | 0.998                          | 0.931        |
| Auto          | 0.99  | 0.944                          | 0.940        | 0.0%  | 0.525                          | <b>0.422</b> |
| RAF           | 0.997 | 0.926                          | 0.917        | 0.0%  | 0.085                          | <b>0.106</b> |

#### Leverage is an upper bound, not a prediction.

Table 6 reports the diagnostic. Read as Corollary 1 intends, it does not say which partition wins; it says how much any partition could win by. Two panels have almost no room on the size block: Auto and RAF retain median  $\lambda^{(+)} \approx 0.92$ –0.94 even at their selected discounts, with no item below 0.1, so their sub-percent pooling differences are explained before any partition is fitted. The remaining three retain leverage: Carparts is extreme—median 0.224, with 22.5% of items essentially fully shrunk—and Online Retail sits at 0.553 under the selected discount. Leverage alone does not fix the prize, however. The controlled study shows that the true partition’s return at a given  $\lambda^{(+)}$  depends on the forgetting regime that produced it and on how far the groups are apart, so the panels are placed on the full surface below.

#### Where the room is, and whether it is collected.

Figure 6 places the five panels on a surface that varies leverage and group separation together. Each panel’s size-block separation is measured as the share of item log-size variance explained by a partition,  $R^2$ , under its learned partition (0.44 to 0.95) and under the objective-free ADI/CV<sup>2</sup> taxonomy (0.01 to 0.30); the simulated cells carry the same statistic under their true labels. Read within each panel’s own forgetting regime (Table 29), a supplied true partition would earn between −0.5% and +0.6% on Online Retail, M5, Auto, and RAF, and 1.7 to 2.0%

on Carparts. The learned partitions return −0.05%, +0.58%, −0.31%, +0.03%, and −0.08%: within the surface’s range on the four panels without room, and about two points short on the one panel with it. At Carparts’s coordinates the simulated learned mixture also loses (−0.7% to −0.8%), and across the grid it is negative at every moderate separation (Figure 6, right), recovering the true partition’s return only when centers are two or more log-units apart and histories are long. The earlier reading of Online Retail as the decisive case rested on an envelope that pooled separations; resolved, its room is at most 0.6%.

This separates the two explanations that aggregate results cannot. On four panels the credibility weights and the separation together say there is little to exploit, and the estimator’s near-zero refinement gains are the predicted outcome, not a failure. On Carparts the structure would pay and the objective does not find it; that the same objective fails on simulated panels at the same coordinates makes learnability, not the bound, the binding constraint there. Corollary 1 bounds the prize; the surface says where the prize is nonzero.

#### High leverage is an amplifier with no sign attached.

The clearest illustration is RAF, which combines the highest size-block leverage in the panel set ( $\lambda^{(+)} = 0.917$ ) with the lowest occurrence leverage ( $\lambda^{(o)} = 0.106$ ), and loses 31.5% of external MAE to a learned partition. The two facts are one fact. The point forecast is  $\hat{y}_i = \hat{\pi}_i \exp(\hat{\mu}_i + \hat{\sigma}_i^2/2)$ ; when 89% of  $\hat{\pi}_i$  comes from the group prior, replacing the group is close to replacing the forecast. This is the empirical content of the caveat attached to Proposition 1: persistent prior leverage guarantees that the choice of target keeps mattering, not that a particular target is good. Read together with the previous paragraph, low leverage caps the upside and high leverage uncaps the downside, and neither makes a learned partition worth using here.

#### Forgetting is an order of magnitude larger.

Computed within a single selection surface under one definition—the M5 surface at its selected discount—refining the pooling structure is worth +0.58% of validation SPL, while discounting at the same structure is worth 8.95% for the global pool and 9.46% for the mixture. The asymmetry that motivates the paper survives at an order of magnitude, and it survives on the one panel where refinement is measurable at all.

Table 7: Online Retail distributional ablation. Each row is an independent hierarchical run; “auto” denotes its own internal discount selection. The raw predictive law is used and lower is better. Full point metrics are in Appendix Table 9.

| Pooling  | Forget     | SPL           | SPL <sub>90</sub> | AIW         |
|----------|------------|---------------|-------------------|-------------|
| global   | off        | 0.8899        | 1.5306            | 10.51       |
| taxonomy | off        | 0.8897        | 1.5299            | 10.56       |
| learned  | off        | 0.8905        | 1.5336            | 10.79       |
| global   | auto (.95) | <u>0.8849</u> | <u>1.5268</u>     | <b>9.25</b> |
| taxonomy | auto (.95) | <b>0.8846</b> | <b>1.5254</b>     | 9.43        |
| learned  | auto (.95) | 0.8869        | 1.5343            | 10.69       |

**What the ablation confirms.** Table 7 is the external counterpart on Online Retail: the three pooling candidates differ by less than 0.1% in mean SPL without forgetting and by less than 0.3% at the selected discount, matching the internal picture. Discounting shows up elsewhere, narrowing the global and taxonomy 80% intervals by 11–12% while improving mean SPL by about half a percent; the full ablation (Appendix Table 9) shows it improves external MAE by 3.5–4.0% and slightly worsens RMSE. The cross-panel result in Table 2 should therefore be attributed to the shared hurdle-EB predictive law and to the temporal half of the selection, not to a learned-partition gain. That is not a concession made after the fact: the leverage diagnostic states the ceiling in advance, and Figure 5 shows we do not reach it.

## 6 Limitations

**(1) The size law.** The log-normal positive-size block keeps item-posterior updates analytic, but its tail is conservative on panels with heavy-tailed spikes. Official TweedieGP is better on Online Retail, Carparts, and M5, and throughout the Carparts far tail; conformal wrappers are better at q90 on M5. Two model-side remedies were tested and neither closes the gap (Appendix E.4): replacing the log-normal size law by log- $t$ , moment-matched Gamma, or inflated-scale variants of the same fitted posterior moves mean SPL by at most 0.003 on every panel, and item-specific forgetting rates with empirical-Bayes shrinkage improve M5 by 1.2% but leave it 1.8% behind TweedieGP and change no other panel by more than 0.1%, and a separate occurrence discount, chosen by the same validation rule, moves M5 by 0.14%. The remaining gap therefore lies in the posterior—per-series temporal adaptation and a single latent process for occurrence and size—rather than in the shape of the predictive

law or the forgetting schedule.

**(2) Saturation and calibration.** Raw intervals over-cover on the sparsest panels (Table 25), and on RAF the standard quantile levels saturate: the all-zero forecast attains a marginally lower mean SPL than every method. The far-tail levels ( $q \geq 0.95$ ) resolve this (Appendix E.14), but CRPS and a cost-based criterion remain unrun.

**(3) Assumptions behind the resolution condition.** Corollary 1 uses a Gaussian working model for the item means and a  $\chi^2$  approximation for  $S_g^2$ , accurate except for small groups combining strongly unequal counts with strongly unequal sampling variances, where it understates the collapse probability by up to 0.21. It is an estimator-specific estimability statement, not an information-theoretic impossibility result. Section 5.6 shows the practical consequence of that distinction: the bound is an upper envelope, and on these panels it is not the binding constraint.

**(4) The learnability gap is demonstrated, not closed.** Our strongest negative result—on the one panel with room, Carparts, neither the real nor the simulated learned partition collects it, and in simulation the mixture objective is negative at every moderate separation and neither label sample-splitting nor credibility shrinkage of the group variance recovers it (Appendix E.8)—identifies a target rather than hitting it. Two extensions are directly implicated and neither is bounded by Corollary 1 in the same way. A soft-responsibility predictive mixture adds one matrix multiply and changes the *shape* of the predictive distribution rather than only the prior it shrinks toward, although the predictive-law variants of Appendix E.4 indicate that shape alone is not where the remaining gap lies. A partition objective that maximizes between-group separation subject to keeping  $\rho_g \sqrt{m_g}$  above threshold would target learnability directly rather than likelihood. We report the gap because we can measure it, not because we have closed it.

**(5) Baseline and inference scope.** We do not run external hierarchical Bayesian implementations [2, 18, 20]; claims about them are limited to their published computational profiles. Appendix E.15 closes the inference-method half of this gap—a four-chain full-posterior Gibbs sampler matches plug-in EB within 0.1% on SPL—so the EB approximation is not the distributional-accuracy bottleneck, but the model-class half remains open. Our adaptive conformal comparison is one ACI design, not an exhaustive study of weighted or model-specific conformal methods; given how much it changed the baseline (Section 5.3), the remaining designs deserve more attention than they have received in this literature.

DeepState, Chronos-Bolt, and Deep Renewal Process comparisons use subset or proxy protocols and are reported only in Appendix E.13.

**(6) Criterion and deployment alignment.** Selection uses a single pinball-based score; the selector tracks whichever criterion it is given (Section E.6), but a cross-objective study including an inventory cost criterion is future work. The discount is selected under the fixed-origin protocol and reused in walk-forward: we keep the initialization discount and the online block update distinct and do not compound them, and we do not evaluate re-selection at every block. Appendix E records the remaining protocol details, including the reduced refit cadence for AutoARIMA, AutoTheta, and iETS on Online Retail.

## 7 Conclusion

Intermittent-demand forecasting requires two structural choices: how strongly to discount the past, and how to pool evidence across items. EBB makes both by chronological validation inside a scalable sufficient-statistic empirical-Bayes family, applying one recency operator to item- and group-level statistics alike. Across five public panels and two protocols it is within 3% (fixed origin) and 5% (walk-forward) of the best competing method everywhere, a property shared only by an official Tweedie–Gaussian-process specialist that costs one to two orders of magnitude more and must be refit at every update. EBB leads that specialist on Auto and RAF and trails it by under one percent on Online Retail and Carparts and by 2.9% on M5 at fixed origin. Under walk-forward updating the adaptive conformal layer decides M5: applied to EBB it leads, applied to ADIDA it is second, and uncalibrated EBB is third; a validation-tail rule selects the layer where it helps and leaves EBB within 0.8% of the best competing method on every panel. These are distributional results obtained from a closed-form, CPU-only, bit-reproducible predictive law; point forecasts are competitive but not uniformly best.

The two choices turn out to be asymmetric, and the asymmetry is the paper’s finding. Discounting is a first-order decision: all five panels select it, and on the one panel where both effects are measurable it is worth an order of magnitude more than refining the pooling structure. Refinement, by contrast, improves external accuracy on none of the five panels and exceeds selector noise on only one.

Why it fails is the more useful result. Forgetting has two opposing effects on pooling—it keeps a shared prior’s leverage alive while eroding the infor-

mation that would resolve fine group structure—and the resulting credibility weight bounds how much any partition could be worth. Once group separation is placed on the same surface as leverage, that bound binds on four of five panels: a supplied true partition would earn nothing there, and nothing is found. On Carparts it would earn 1.8 to 2.4%, the learned partition returns  $-0.08\%$ , and the same objective loses at those coordinates in simulation. The constraint on cross-series pooling in these data is therefore the absence of room on most panels and learnability on the one panel that has it. Separating those two explanations required the diagnostic, and the diagnostic costs nothing beyond the fit that produces it.

Two evaluation conventions actively obscure this picture, and both are inherited rather than incidental. Absolute-error metrics reward the all-zero forecast on three of five panels and reverse the ranking of a known-correct partition in simulation; they should not be the sole basis for comparing intermittent-demand forecasters. Static split-conformal wrappers, the field’s standard probabilistic baseline, understate what conformal prediction achieves here by roughly twenty percent relative to an adaptive construction—a margin several times larger than most method differences reported on these panels, including our own.

The practical value of EBB is accordingly not a claim of universal accuracy but a cheap, deterministic predictive distribution together with an ex ante account of when temporal adaptation and finer pooling can matter. The most direct next steps follow from the gap the diagnostic exposes: partition objectives that optimize resolvable separation rather than marginal likelihood, predictive mixtures that alter distributional shape rather than only the shrinkage target, and decision-aware sequential selection.

## Reproducibility Statement

All five datasets are public (UCI 352; Kaggle M5-Accuracy; the gluon-ts `intermittent-datasets` branch for Auto and RAF; Zenodo record 4656022 for Carparts). Conversion and preprocessing scripts are included in the released code. The full pipeline uses deterministic low-dimensional EB optimization and seed-free quantile-based EM initialization, so no seed policy is required for the forecaster itself. The stochastic elements are the seed-controlled M5 subsample and the  $B=20$  bootstrap; M5 results are reported for seed 42 in the main tables and audited across three independently drawn samples, which select the same configuration in every case. An au-

tomated test asserts that two end-to-end runs are bit-identical. Every table maps to one experiment directory via a manifest file, and each run writes its selection diagnostics—chosen partition, discount, BIC curves, and credibility leverage—alongside its metrics. Hardware and timing methodology are specified in Appendix D. The complete experiment suite runs on one CPU workstation in under a day, dominated by the statistical baselines rather than EBB. An anonymized code repository will accompany the submission.

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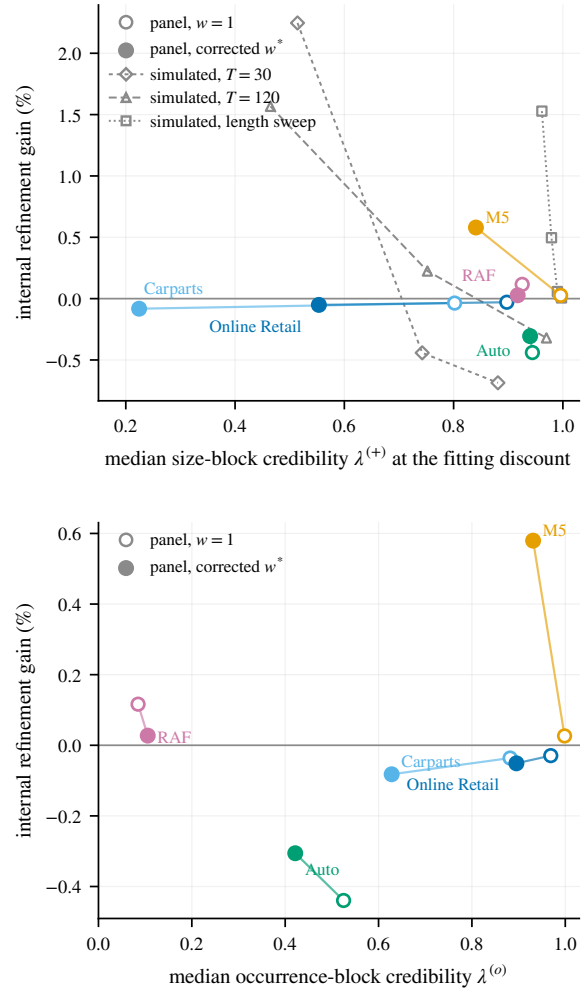


Figure 5: **The simulated curves are an upper envelope that the real panels do not reach.** Internal refinement gain (best refined structure versus the global pool, validation SPL) against median credibility weight, on the size block (top) and the occurrence block (bottom). Open markers are  $w=1$ , filled markers the selected discount, and the connecting segment is the movement forgetting induces. Grey curves are the controlled panels of Section 5.5, where the partition supplied is correct by construction. The grey curves pool the three separations of the controlled grid; Figure 6 resolves them. Read against the pooled envelope, every real panel lies below it at its own leverage.

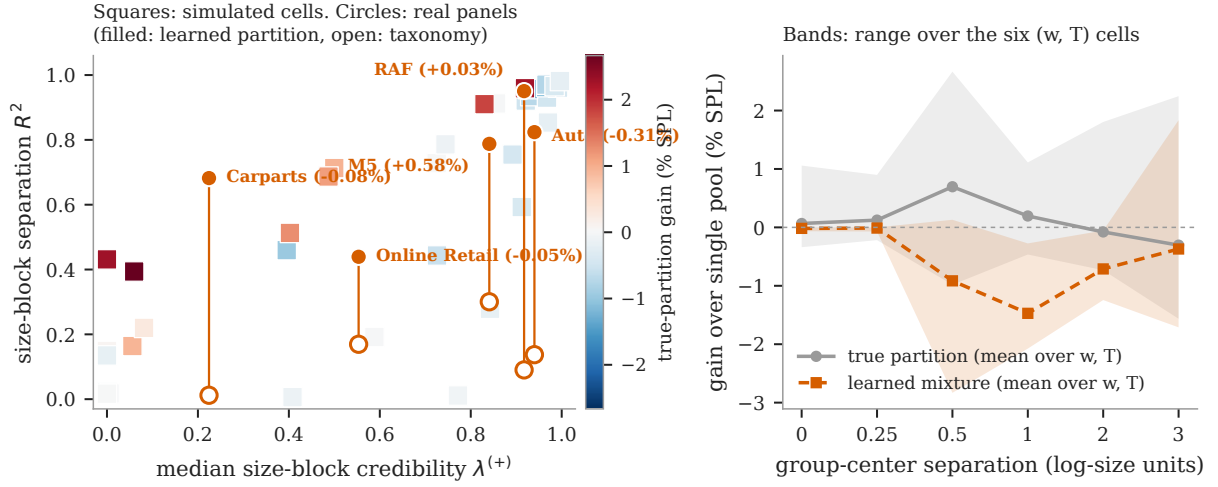


Figure 6: **Room depends on separation as well as leverage, and the real panels have little of it.** Left: each square is one cell of the controlled grid (separation  $\times$  discount  $\times$  length, nine paired replicates; occurrence separation scaled with size separation), placed at its median size-block credibility and at the share of item log-size variance its true partition explains ( $R^2$ ), colored by the true partition's SPL gain over a single pool. Real panels are placed at the same two coordinates, with  $R^2$  measured under the learned partition (filled) and under the objective-free ADI/CV<sup>2</sup> taxonomy (open); labels give the learned partition's realized refinement gain. Right: the true partition's gain and the learned mixture's gain against separation, averaged over the six ( $w, T$ ) cells, with the range across cells shaded. Room read within each panel's own forgetting regime is in Table 29.

## A Notation

|                                              |                                                                   |
|----------------------------------------------|-------------------------------------------------------------------|
| $y_{i,t}, z_{i,t}, \ell_{i,t}$               | demand, occurrence indicator, log-size                            |
| $T_i$                                        | initialization-window length of item $i$                          |
| $g(i), K$                                    | pooling group of item $i$ ; number of mixture components          |
| $\alpha_g, \beta_g$                          | Beta occurrence prior; $\phi_g = \alpha_g + \beta_g$ its strength |
| $\mu_{0,g}, \tau_g^2, \sigma_g^2$            | size prior location, between/within-item variances                |
| $w; b$                                       | recency discount; forgetting strength $b = 1 - w$                 |
| $n_i(w), c_i(w)$                             | discounted observation / occurrence pseudo-counts                 |
| $\bar{p}_i(w)$                               | discounted empirical occurrence rate $c_i(w)/n_i(w)$              |
| $\lambda_i(w)$                               | credibility weight $n_i(w)/(n_i(w) + \phi_g)$                     |
| $\hat{\pi}_i, \hat{\mu}_i, \hat{\sigma}_i^2$ | posterior means of occurrence, log-size, variance                 |
| $\alpha_p$                                   | classical TSB occurrence smoothing constant ( $= 1 - w$ )         |
| $\delta$                                     | occurrence drift rate (Proposition 4)                             |
| $M, n$                                       | candidate count and validation size (Prop. 5)                     |
| $B$                                          | bootstrap draws for hyperparameter averaging                      |

## B Supporting Results

This appendix records four connections that support the construction but are not the paper’s main theoretical contribution: the relation to TSB, a tracking bound under drift, finite-class selection, and classical credibility.

### B.1 Relation to the TSB occurrence recursion

Write  $p_t = z_{i,t}$  for one fixed item. Let  $\hat{p}_T$  be classical TSB’s occurrence estimate with smoothing  $\alpha_p = 1 - w$  and initialization  $\hat{p}_0$ .

**Proposition 3** (TSB occurrence as a boundary case). For  $w \in (0, 1)$ : (i)  $\lim_{(\alpha, \beta) \rightarrow (0, 0)} \hat{\pi}(w) = \bar{p}(w)$ ; (ii)  $\hat{p}_T = (1 - w) \sum_{t=1}^T w^{T-t} p_t + w^T \hat{p}_0$  and  $\bar{p}(w) = (\hat{p}_T - w^T \hat{p}_0)/(1 - w^T)$ ; (iii) hence  $\bar{p}(w) = \hat{p}_T$  under the self-consistent initialization  $\hat{p}_0 = \bar{p}(w)$ , and  $|\bar{p}(w) - \hat{p}_T| \leq w^T$  in general.

The equivalence concerns only the occurrence block. Classical TSB uses arithmetic smoothing for positive sizes, whereas EBB discounts positive log-sizes under (1).

### B.2 Tracking bound under occurrence drift

**Proposition 4** (Tracking risk under drift). If  $p_t \sim \text{Bern}(\pi_t)$  independently and  $|\pi_t - \pi_T| \leq \delta(T - t)$ , then  $|\mathbb{E}\bar{p}(w) - \pi_T| \leq \delta w/[(1 - w)(1 - w^T)]$  and  $\text{Var}\bar{p}(w) \leq \frac{1}{4}(1 - w)(1 + w^T)/[(1 + w)(1 - w^T)]$ , with the corresponding bias-variance decomposition for the posterior mean.

**Corollary 2.** Balancing the two bounds gives  $1 - w^* \asymp \delta^{2/3}$ , worst-case MSE  $O(\delta^{2/3})$ , and  $w^* \rightarrow 1$  as  $\delta \rightarrow 0$ .

This result motivates including  $w = 1$  in the candidate grid; it does not claim that chronological validation estimates the bound-optimal  $w^*$ .

### B.3 Finite candidate selection

**Proposition 5** (Conditional finite-class bound). Let  $R(c)$  be the expected validation risk of candidate  $c$  and  $\bar{R}(c)$  its average over  $n$  validation items. If the item losses are independent conditional on the fitted candidates and lie in  $[0, B]$ , then the minimizer  $\hat{c}$  over  $M$  candidates satisfies, with probability at least  $1 - \eta$ ,

$$R(\hat{c}) \leq \min_c R(c) + B \sqrt{\frac{2 \log(2M/\eta)}{n}}.$$

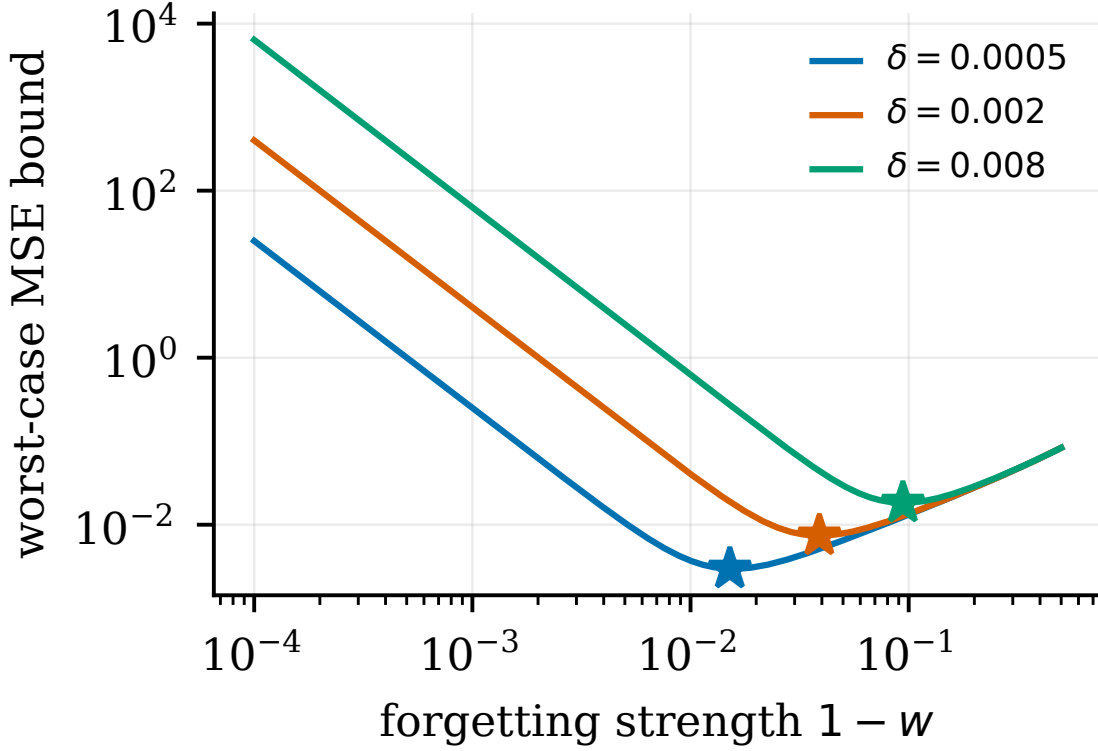


Figure 7: Bias–variance bounds in Proposition 4. Stars mark the minimizing forgetting strength; as drift  $\delta$  increases, the optimum moves at the rate in Corollary 2.

The independent units are items, because observations and quantiles are averaged within item before selection. The released selector does not clip scaled pinball loss, so the bounded-loss assumption is not automatic and this proposition is not presented as a formal guarantee for the unclipped implementation. It records the standard finite-class behavior under an explicit clipping condition.

## B.4 Classical credibility

**Proposition 6** (Pooling reduces Bayes risk in the stationary model). Under the undiscounted Beta–Binomial occurrence model with known hyperparameters, the posterior mean is the Bayes estimator of  $\pi_i$  under squared loss and has strictly smaller Bayes risk than the per-item empirical rate whenever  $\phi_g > 0$ . Its optimal weight is the classical credibility factor  $\lambda_i = n_i / (n_i + \phi_g)$  [1].

This is the Bühlmann–Straub result specialized to Bernoulli occurrences. It justifies pooling when the stationary hierarchical model is correctly specified; it does not assert that a misspecified or stale group prior helps under drift.

## C Proofs

### C.1 Proof of Proposition 3

(i) Immediate from (6) since  $c(w), n(w)$  are finite. (ii) Induction:  $\hat{p}_1 = (1 - w)p_1 + w\hat{p}_0$ . If the closed form holds at  $T - 1$ , then  $\hat{p}_T = (1 - w)p_T + w\hat{p}_{T-1} = (1 - w) \sum_{t=1}^T w^{T-t} p_t + w^T \hat{p}_0$ . With  $n(w) = (1 - w^T)/(1 - w)$ ,

$$\bar{p}(w) = \frac{(1 - w) \sum_{t=1}^T w^{T-t} p_t}{1 - w^T} = \frac{\hat{p}_T - w^T \hat{p}_0}{1 - w^T}.$$

(iii) Rearranging,  $\bar{p}(w) - \hat{p}_T = w^T(\bar{p}(w) - \hat{p}_0)$ , and  $|\bar{p}(w) - \hat{p}_0| \leq 1$ .  $\square$

## C.2 Proof of Proposition 1

$n(w) = (1 - w^T)/(1 - w) \leq 1/(1 - w)$  and  $x \mapsto x/(x + \phi)$  is increasing, so  $\lambda(w) \leq [1/(1 - w)]/[1/(1 - w) + \phi] = 1/(1 + (1 - w)\phi)$ .  $\square$

## C.3 Proof of Proposition 4 and Corollary 2

With  $k = T - t$  and  $\sum_{k \geq 0} kw^k = w/(1 - w)^2$ ,

$$|\mathbb{E}\bar{p}(w) - \pi_T| = \left| \frac{\sum_t w^{T-t}(\pi_t - \pi_T)}{n(w)} \right| \leq \frac{\delta w/(1 - w)^2}{(1 - w^T)/(1 - w)} = \delta \frac{w}{(1 - w)(1 - w^T)}.$$

By independence and  $\pi_t(1 - \pi_t) \leq \frac{1}{4}$ ,

$$\text{Var } \bar{p}(w) \leq \frac{\frac{1}{4}(1 - w^{2T})/(1 - w^2)}{[(1 - w^T)/(1 - w)]^2} = \frac{1}{4} \cdot \frac{(1 - w)(1 + w^T)}{(1 + w)(1 - w^T)}.$$

Decomposing  $\hat{\pi} - \pi_T = \lambda(\bar{p} - \mathbb{E}\bar{p}) + \lambda(\mathbb{E}\bar{p} - \pi_T) + (1 - \lambda)(\mu_\pi - \pi_T)$ , the first term is zero-mean and independent of the deterministic remainder, giving  $\mathbb{E}(\hat{\pi} - \pi_T)^2 \leq [\lambda B(w) + (1 - \lambda)|\mu_\pi - \pi_T|]^2 + \lambda^2 \text{Var } \bar{p}(w)$  with  $B(w)$  the bias bound. For the corollary set  $b = 1 - w$  and let  $T \rightarrow \infty$  (so  $w^T \rightarrow 0$ ): the squared bias bound behaves as  $\delta^2(1 - b)^2/b^2 \sim \delta^2/b^2$  and the variance bound becomes  $\frac{1}{4}b/(2 - b)$ , which is  $b/8$  to leading order as  $b \rightarrow 0$ . Minimizing  $f(b) = \delta^2/b^2 + b/8$  gives  $f'(b) = -2\delta^2/b^3 + 1/8 = 0$ , i.e.,  $b^* = (16\delta^2)^{1/3} = 2^{4/3}\delta^{2/3}$ , and  $f(b^*) = O(\delta^{2/3})$ .  $\square$

## C.4 Proof of Proposition 5

For each candidate  $c$ , Hoeffding's inequality for independent scores in  $[0, B]$  gives  $\Pr(|\hat{R}(c) - R(c)| > t) \leq 2\exp(-2nt^2/B^2)$ . A union bound over the  $M$  candidates makes  $\sup_c |\hat{R}(c) - R(c)| \leq t$  hold with probability at least  $1 - 2M\exp(-2nt^2/B^2)$ . Setting the right side to  $\eta$  yields  $t = B\sqrt{\log(2M/\eta)/(2n)}$ . On this event,  $R(\hat{c}) \leq \hat{R}(\hat{c}) + t \leq \hat{R}(c^*) + t \leq R(c^*) + 2t$  with  $c^* = \arg \min_c R(c)$ , which is the claim with constant  $2t = B\sqrt{2\log(2M/\eta)/n}$ .  $\square$

## C.5 Proof of Proposition 6

With known  $(\alpha_g, \beta_g)$  and  $s_i \mid \pi_i \sim \text{Bin}(n_i, \pi_i)$ ,  $\pi_i \sim \text{Beta}(\alpha_g, \beta_g)$ , the posterior mean  $\hat{\pi}_i = \lambda_i \bar{p}_i + (1 - \lambda_i)\mu_\pi$  with  $\lambda_i = n_i/(n_i + \phi_g)$  minimizes Bayes risk under squared loss among all estimators (tower property). Restricting to linear combinations  $a\bar{p}_i + b$ , the Bayes risk is  $a^2 \mathbb{E}[\text{Var}(\bar{p}_i \mid \pi_i)] + (1 - a)^2 \text{Var}(\pi_i)$  at the optimal intercept, minimized at  $a^* = \text{Var}(\pi_i)/(\text{Var}(\pi_i) + \mathbb{E}[\text{Var}(\bar{p}_i \mid \pi_i)]) = \lambda_i$ , the Bühlmann credibility weight. The minimum is strictly below the risk of  $a = 1$  (the empirical rate) whenever  $\text{Var}(\pi_i) < \infty$  and  $\phi_g > 0$ .  $\square$

## C.6 Proof of Proposition 2 and Corollary 1

(i)  $\lambda_i = n_i/(n_i + \kappa_g)$  is increasing in  $n_i$ , and  $n_i \leq \sum_{k \geq 0} w^k = (1 - w)^{-1}$  for  $w < 1$ . Substituting the bound,  $\lambda_i \leq \frac{(1-w)^{-1}}{(1-w)^{-1} + \kappa_g} = [1 + (1 - w)\kappa_g]^{-1}$ , which is  $< 1$  whenever  $w < 1$  and  $\kappa_g > 0$ .

(ii) Write  $\bar{y}_i = \mu_i + e_i$  with  $\mu_i \sim N(\mu_g, \tau_g^2)$  and  $e_i \sim N(0, s_i)$  independent,  $s_i = \sigma_g^2/n_i$ , so  $\text{Var}(\bar{y}_i) = \tau_g^2 + s_i$ . For the sample variance  $S_g^2 = (m_g - 1)^{-1} \sum_{i \in g} (\bar{y}_i - \bar{\bar{y}})^2$ , independence gives  $\text{Var}(\bar{\bar{y}}) = m_g^{-2} \sum_i (\tau_g^2 + s_i)$  and hence

$$\mathbb{E}[S_g^2] = \frac{\sum_i (\tau_g^2 + s_i) - m_g \text{Var}(\bar{\bar{y}})}{m_g - 1} = \frac{(1 - m_g^{-1}) \sum_i (\tau_g^2 + s_i)}{m_g - 1} = \tau_g^2 + \bar{s}_g,$$

exactly, with no balance assumption. The estimator truncates precisely when  $S_g^2 \leq \bar{s}_g$ . Under the working model  $S_g^2$  is approximately  $(\tau_g^2 + \bar{s}_g)\chi_{m_g-1}^2/(m_g - 1)$ , so  $\text{Var}(S_g^2) \simeq \frac{2}{m_g-1}(\tau_g^2 + \bar{s}_g)^2$  and the standardized

distance from the truncation boundary is

$$\frac{\mathbb{E}[S_g^2] - \bar{s}_g}{\sqrt{\text{Var}(S_g^2)}} = \frac{\tau_g^2}{\sqrt{2/(m_g - 1)}(\tau_g^2 + \bar{s}_g)} = \sqrt{\frac{m_g - 1}{2}} \rho_g,$$

giving the stated normal approximation for  $\Pr[S_g^2 \leq \bar{s}_g]$ . The discount bound follows from  $n_i \leq (1 - w)^{-1}$ :  $s_i = \sigma_g^2/n_i \geq (1 - w)\sigma_g^2$  for every  $i$ , hence  $\bar{s}_g \geq (1 - w)\sigma_g^2$ , and  $\rho_g$  is decreasing in  $\bar{s}_g$ . Finally  $\hat{\tau}_g^2 = 0$  gives  $\kappa_g = \infty$  and  $\lambda_i = 0$ .

**Corollary.** Requiring the collapse probability to stay below a fixed level is, by (ii), the condition  $\rho_g \sqrt{(m_g - 1)/2} \geq c$  for a constant  $c$  of order one. Solving  $\rho_g = \tau_g^2/(\tau_g^2 + \bar{s}_g) \geq c\sqrt{2/(m_g - 1)}$  for  $\tau_g^2$  gives  $\tau_g^2 \geq \bar{s}_g c\sqrt{2}/(\sqrt{m_g - 1} - c\sqrt{2})$ , i.e. the stated order  $\bar{s}_g/(\sqrt{m_g - 1} - 1)$  up to the constant. Refining a partition decreases  $m_g$  and, by construction of the mixture objective,  $\tau_g^2$ , while decreasing  $w$  raises the floor  $(1 - w)\sigma_g^2$  on  $\bar{s}_g$ . Each shrinks the left side or raises the right side.  $\square$

The algebraic steps, the exact unbiasedness in (ii), and the monotonicity claims are verified with a computer algebra system in the released test suite. The  $\chi^2$  approximation in (ii) is checked against Monte Carlo simulation, agreeing to within 0.05 in probability across the regimes we use.

## D Implementation details

**EB fitting.** Group Beta-Binomial hyperparameters maximize the (weighted) marginal likelihood by L-BFGS-B with positivity bounds; the size block uses pooled within-item variance for  $\sigma_g^2$  and an intercept-only restricted marginal likelihood in  $\tau_g^2$  with profiled  $\mu_{0,g}$ ; item-level process variances use the conjugate scaled-inv- $\chi^2$  update with  $\nu=20$ . With discounting, every count is replaced by its weighted analogue (4); the Beta-Binomial marginal extends to real pseudo-counts via log-Gamma functions. The predictive law is zero-inflated log-normal with predictive variance  $\hat{\sigma}_{\text{proc}}^2 + \hat{v}_\mu$ . The released configuration averages quantiles over  $B=20$  bootstrap resamples of the item-statistics rows. An optional location-scale calibration layer exists but is *off* in all headline results; the adaptive conformal comparison is reported as a separate baseline rather than used to modify EBB after fitting.

**Mixture EM.** Deterministic quantile initialization over (occurrence rate, mean log-size); degenerate components (mass  $< 2$ ) frozen to global fallback priors; tolerance  $10^{-5}$ , max 60 iterations;  $K \in \{1, \dots, 6\}$  by BIC with  $5K + (K - 1)$  parameters.

**Selection.** Internal chronological split (first 80% / last 20%, min head 8); score = mean pinball over  $q \in \{0.5, 0.75, 0.9\}$ , per-series scaled by head mean absolute demand, averaged within item and then uniformly across items; grid over  $w \in \{1, 0.999, 0.997, 0.995, 0.99, 0.98, 0.95, 0.90\}$  and partitions {global, taxonomy, mixture(BIC)}. Selected configurations per dataset are reported beneath Table 8; diagnostics (`remix_selection_diagnostics.csv`) accompany every run.

**Environment and timing methodology.** All experiments ran on one Windows 11 workstation (Intel Core Ultra 9 275HX, 32 GB RAM). Core determinism tests are bit-identical within this environment; cross-environment bit identity is not claimed. Timing is wall-clock and excludes data loading. Joint selection over the 24 candidates takes 51–243 seconds per panel, with one deterministic six- $K$  EM/BIC search for the mixture on the fitting head. Final model-only point plus five-quantile prediction with  $B = 20$  takes 6.2 seconds on Carparts, 38.0 on Online Retail, and 124.0 on M5, whose external span has 3.235 million observations. GPU is not used in EBB.

**Baselines.** StatsForecast defaults; TSB  $(\alpha_d, \alpha_p) = (0.5, 0.45)$ ; AutoARIMA/AutoTheta season 7 (daily) / 12 (monthly) for intervals. TSB-tuned:  $5 \times 5$  grid  $\alpha_d \in \{0.05, 0.1, 0.2, 0.35, 0.5\}$ ,  $\alpha_p \in \{0.01, 0.05, 0.1, 0.25, 0.45\}$  (25 candidates, the same order as EBB’s 24), scored on the identical 80/20 chronological split by per-series scaled MAE (so neither method is scored on its home criterion), refit on the full window. Selected  $(\alpha_d, \alpha_p)$ : Online Retail (0.10, 0.25); M5 (0.20, 0.01); Auto (0.20, 0.10); Carparts (0.50, 0.45), i.e. the textbook constants;

Table 8: Point forecasting across the five panels (lower is better; bold best and underline second among non-degenerate methods). Degenerate rows are shown as metric diagnostics and excluded from ranking.

| Model          | Online Retail |               | M5            |               | Auto          |               | Carparts      |               | RAF           |               |
|----------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                | MAE           | RMSSE         | MAE           | RMSSE         | MAE           | RMSSE         | MAE           | RMSSE         | MAE           | RMSSE         |
| CrostonClassic | 6.3294        | 5.1051        | 1.2345        | 2.3539        | 3.4056        | 1.1488        | 0.6792        | 1.0981        | 2.7315        | 1.6280        |
| CrostonSBA     | 6.1953        | 5.0692        | 1.2076        | 2.3517        | <u>3.3325</u> | <u>1.1451</u> | 0.6628        | 1.0935        | 2.6520        | 1.6272        |
| TSB            | <u>5.5736</u> | 4.8031        | 1.2672        | 2.3612        | 3.6587        | 1.2066        | <b>0.5396</b> | 1.0156        | <b>2.1660</b> | 1.6570        |
| TSB-tuned      | 5.6318        | 4.7985        | <b>1.1932</b> | <u>2.3137</u> | 3.3917        | 1.1452        | <b>0.5396</b> | 1.0156        | <u>2.2447</u> | 1.6408        |
| ADIDA          | 5.6860        | <u>4.7967</u> | 1.2274        | 2.3398        | 3.4079        | 1.1545        | 0.5556        | 1.0244        | 2.2977        | <u>1.6245</u> |
| IMAPA          | 5.7000        | 4.7970        | 1.2317        | 2.3280        | 3.4143        | 1.1555        | <u>0.5524</u> | 1.0126        | 2.2850        | 1.6259        |
| AutoARIMA      | 6.5494        | 4.8175        | 1.2283        | 2.3169        | 3.4417        | 1.1745        | 0.5822        | <u>1.0105</u> | 2.3153        | 1.6262        |
| AutoTheta      | 8.2862        | 4.8550        | 1.2988        | 2.3143        | 3.7490        | 1.2215        | 0.5704        | <b>0.9966</b> | 2.3687        | 1.6399        |
| Zero           | 4.7549        | 4.8089        | 1.2995        | 2.4629        | 4.1444        | 1.4454        | 0.3867        | 1.0949        | 1.1717        | 1.6345        |
| Naive-last     | 6.0570        | 4.8324        | 1.3750        | 2.5450        | 4.3529        | 1.3818        | 0.5399        | 1.1408        | 1.8340        | 1.7644        |
| EBB (ours)     | <b>5.5325</b> | <b>4.7901</b> | <u>1.2032</u> | <b>2.2938</b> | <b>3.2145</b> | <b>1.1374</b> | 0.5726        | 1.0120        | 2.3874        | <b>1.6245</b> |

RAF (0.50, 0.25). Static conformal wrappers use a chronological 80/20 split and a *panel-pooled* signed-residual distribution; quantiles are zero-truncated and monotonized. The ACI variants retain that residual pool and adapt the requested level for every series-quantile pair with  $\gamma = 0.05$ . Tweedie-GLM is a per-series autoregressive GLM with log link,  $p=1.5$ , 14 lags (daily) / 6 (monthly), and  $\ell_2=0.1$ . TweedieGP uses the authors’ official implementation with Tweedie likelihood, median-demand scaling, RBF kernel, and 50,000 predictive samples; zero-only training series receive a zero fallback so that every series is scored. iETS uses `smooth::adam` with automatic occurrence modeling.

## E Additional results

### E.1 Point forecasting and horizon decomposition

Table 8 gives the complete point comparison retained for continuity with the intermittent-demand literature. Rankings exclude the deliberately degenerate Zero and Naive-last rows. EBB takes six of ten MAE/RMSSE cells and is top two on seven, but the result is not uniform: classical methods retain several point-metric wins, and Zero attains the lowest MAE on Online Retail, Carparts, and RAF. This is why the main text uses point accuracy as a secondary diagnostic.

The selected  $(g, w)$  pairs are: Online Retail (global, 0.95), M5 (mixture, 0.95), Auto (global, 0.99), Carparts (global, 0.90), and RAF (mixture, 0.997). On three of the five panels the margins between pooling structures on the internal validation surface are under a tenth of a percent, so the structure component of these pairs should be read as Section 5.6 reads it: the selector avoids poor structural corners, and no winner is announced where the surface does not resolve one.

### E.2 Selector diagnostics

The selection surface is locally flat near its optimum: on Online Retail the best seven candidates span all three pooling structures and lie within 0.51% of one another in validation SPL. Section 5.6 reports the credibility leverage and refinement gains that accompany each selected pair, and Figure 5 places them against the controlled panels. The selector is useful for choosing a discount and for avoiding poor structural corners; the winning pooling structure on a panel whose candidates lie within a tenth of a percent should not be read as a scientific taxonomy. Per-panel selection surfaces are released with the code.

### E.3 Walk-forward point results and protocol

EBB applies the selected discount once to the initialization sufficient statistics and then accumulates revealed blocks without compounding it. On Online Retail, ACI-ADIDA improves the static conformal wrapper

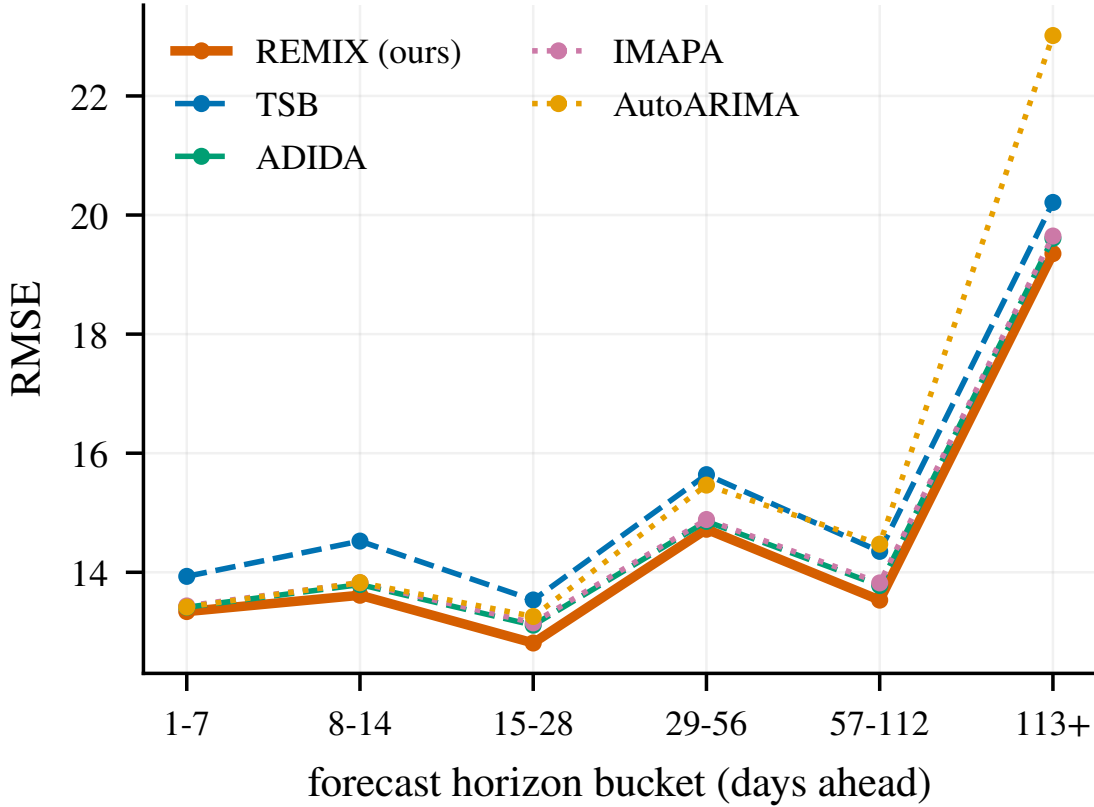


Figure 8: Online Retail RMSE by fixed-origin forecast-horizon bucket. EBB is lowest in every bucket, so the aggregate result is not created by averaging only over distant horizons. MAE differences among the leading methods are within 1–2% by bucket.

substantially and wins q50 by a hair; TweedieGP takes the mean and q90, and EBB takes q75 and is second elsewhere. On Carparts the first two quantiles saturate at zero, EBB leads from q10 through q75, and TweedieGP takes q90 and the mean. On Auto EBB leads q10, q25, and the mean while TweedieGP leads q50 through q90; on RAF EBB leads every quantile, coinciding with the zero forecast through q75; on M5 ACI-EBB leads every quantile from q25 up and the mean, ACI-ADIDA is second, and uncalibrated EBB leads q10. ACI-EBB improves on EBB on M5, Auto, and Carparts and degrades it on Online Retail and RAF (Table 22). Table 16 gives the per-panel regrets behind Table 4. Re-running joint selection at every block is not evaluated.

#### E.4 Three model-side remedies that do not close the gap

Section 5.2 locates the remaining shortfall to TweedieGP on Online Retail, Carparts, and M5. Three remedies within the EBB family are the obvious candidates: a heavier-tailed positive-size law, an item-specific forgetting rate, and a separate forgetting rate for the occurrence block. Both were evaluated on the fixed-origin protocol with the audited pooling structure and, for the size law, the identical fitted posterior; neither changes any panel’s leader.

**Predictive-law shape.** Table 18 replaces the log-normal predictive law of (3) by log- $t$ , moment-matched Gamma, and inflated-scale alternatives computed from the same item posteriors. Mean SPL moves by at most 0.003 on every panel and in no case reaches TweedieGP.

**Item-specific forgetting.** Each item receives its own discount  $w_i$ , chosen on an inner chronological split of the fitting head by minimizing  $n_i^V R_i(w) + \kappa \bar{R}(w)$ , where  $R_i$  is the item’s validation pinball curve over the

Table 9: Full Online Retail ablation corresponding to Table 7. Each row is an independent run; “auto” denotes its own internal discount selection. TSB has no native predictive distribution. Lower is better.

| Pooling    | Forget     | MAE           | RMSE           | RMSSE         | SPL           | SPL <sub>90</sub> | AIW         |
|------------|------------|---------------|----------------|---------------|---------------|-------------------|-------------|
| none (TSB) | fixed      | 5.5736        | 18.7272        | 4.8031        | —             | —                 | —           |
| global     | off        | 5.7623        | <b>17.6892</b> | <b>4.7851</b> | 0.8899        | 1.5306            | 10.51       |
| taxonomy   | off        | 5.7663        | <u>17.6930</u> | <u>4.7875</u> | 0.8897        | 1.5299            | 10.56       |
| learned    | off        | 5.8126        | 17.7064        | 4.7876        | 0.8905        | 1.5336            | 10.79       |
| global     | auto (.95) | <b>5.5325</b> | 17.8386        | 4.7901        | <u>0.8849</u> | <u>1.5268</u>     | <b>9.25</b> |
| taxonomy   | auto (.95) | <u>5.5624</u> | 17.8421        | 4.7897        | <b>0.8846</b> | <b>1.5254</b>     | 9.43        |
| learned    | auto (.95) | 5.6925        | 17.8765        | 4.7902        | 0.8869        | 1.5343            | 10.69       |

Table 10: Online Retail point forecasting under strict seven-day walk-forward updates. The hierarchical rows differ in their initialization discount and pooling structure.

| Model                            | MAE          | RMSE          | RMSSE        |
|----------------------------------|--------------|---------------|--------------|
| CrostonClassic                   | 5.692        | 16.937        | 4.854        |
| CrostonSBA                       | 5.586        | 16.931        | 4.836        |
| TSB                              | 5.434        | 17.754        | 4.815        |
| ADIDA                            | <b>5.320</b> | <b>16.728</b> | <u>4.637</u> |
| IMAPA                            | <u>5.332</u> | <u>16.745</u> | <b>4.632</b> |
| AutoARIMA                        | 5.482        | 16.904        | 5.105        |
| AutoTheta                        | 5.394        | 16.836        | 4.735        |
| Stationary hierarchy ( $w = 1$ ) | 5.586        | 17.260        | 4.725        |
| Forgetting hierarchy (tax., .95) | 5.411        | 17.059        | 4.682        |
| EBB (selected global, .95)       | 5.421        | 17.188        | 4.696        |

grid of Section 3.4,  $\bar{R}$  the panel average, and  $\kappa$  an empirical-Bayes shrinkage strength selected on the outer validation tail;  $\kappa=\infty$  recovers the panel-wide  $w^*$  and  $\kappa=0$  leaves each item unshrunk. Table 19 reports the result. The pre-registered adoption criterion (improve Online Retail or M5 by at least 1% without degrading Auto or RAF by more than 0.3%, and either recover Online Retail from TweedieGP or bring M5 within 1.5% of it) fails on its second clause: M5 improves by 1.15% but remains 1.8% behind, and Online Retail is unchanged. Unshrunk selection ( $\kappa=0$ ) is markedly worse on M5, so the shrinkage is necessary; the selected  $\kappa$  is interior on four panels. The per-item option is retained in the released code but is not used in any reported result.

**Separate occurrence forgetting.** The M5 shortfall sits in the occurrence block (Section 5.2), and TSB itself smooths occurrence and size with two constants, so a third remedy gives the occurrence counts their own discount  $w^{(o)}$  while the size statistics keep the selected  $w$ .  $w^{(o)}$  is chosen on the same 80/20 chronological split and criterion as  $w$ , from a grid that contains the tied value. Table 20 reports the result. The selector keeps the tied rate on Online Retail, Auto, and RAF, moves M5 by 0.14%, and moves Carparts by 0.77% (to 0.3204, marginally ahead of TweedieGP); the pre-registered criterion of a 1% gain on M5 is not met, so the single rate is retained. A faster occurrence rate ( $w^{(o)}=0.8$ ) removes every zero q90 forecast on M5 and lowers q90 SPL from 2.476 to 2.452 without changing the mean, which locates the M5 tail problem in occurrence persistence rather than in the size law, but the selector does not choose it and we do not override the selector.

## E.5 Significance and interval diagnostics

Table 21 tests the headline metric directly: per-series mean SPL, EBB against every comparator of Table 2, with paired  $t$  and Wilcoxon signed-rank tests Holm-adjusted within each panel and a comparison counted as decided only when both adjusted tests agree at the 5% level. EBB is significantly better in 43 of the 50 panel-comparator pairs, significantly worse in one (TweedieGP on M5, adjusted  $p = 0.003$  and  $0.035$ ), and undecided in six. On M5 the mean difference favors EBB against all four conformal wrappers but the paired  $t$  does not reach significance after correction (the Wilcoxon strongly rejects symmetry, so the per-series

Table 11: Full Online Retail walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov<sup>+</sup>@80) coverage and average interval width (AIW). CP and ACI wrappers refit each of the 36 seven-day blocks; AutoARIMA, AutoTheta, iETS, and TweedieGP refit every four blocks; EBB updates sufficient statistics each block. The zero row is excluded from ranking.

| Model             | q10          | q25          | q50          | q75          | q90          | Mean         | Cov@80 | Cov <sup>+</sup> @80 | AIW   |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------|----------------------|-------|
| AutoARIMA         | 0.259        | 0.572        | 1.197        | 1.763        | 1.522        | 1.063        | 0.933  | 0.762                | 16.58 |
| AutoTheta         | 0.233        | 0.584        | 1.276        | 1.717        | 1.484        | 1.059        | 0.931  | 0.765                | 16.43 |
| iETS              | 0.236        | 0.505        | 1.302        | 1.562        | 1.940        | 1.109        | 0.598  | 0.166                | 19.74 |
| CP-Croston        | 0.341        | 0.778        | 1.364        | 1.745        | 2.632        | 1.372        | 0.793  | 0.478                | 5.66  |
| CP-SBA            | 0.334        | 0.762        | 1.343        | 1.725        | 2.722        | 1.377        | 0.793  | 0.481                | 5.75  |
| CP-TSB            | 0.451        | 0.899        | 1.289        | 1.385        | 2.842        | 1.373        | 0.795  | 0.468                | 5.71  |
| CP-ADIDA          | 0.277        | 0.627        | 1.050        | <u>1.319</u> | 2.514        | 1.158        | 0.791  | 0.473                | 5.47  |
| CP-IMAPA          | 0.292        | 0.660        | 1.083        | <u>1.329</u> | 2.522        | 1.177        | 0.791  | 0.471                | 5.47  |
| ACI-ADIDA         | 0.186        | 0.468        | <b>0.909</b> | 1.324        | 1.668        | 0.911        | 0.904  | 0.646                | 19.67 |
| ACI-TSB           | 0.189        | 0.482        | 0.948        | 1.384        | 1.710        | 0.943        | 0.903  | 0.641                | 21.21 |
| TweedieGP         | <u>0.183</u> | <u>0.458</u> | 0.917        | 1.326        | <b>1.379</b> | <b>0.853</b> | 0.924  | 0.709                | 13.03 |
| Zero (diagnostic) | 0.183        | 0.458        | 0.916        | 1.374        | 1.649        | 0.916        | 0.740  | 0.000                | 0.00  |
| EBB               | 0.183        | 0.458        | <u>0.910</u> | <b>1.319</b> | <u>1.400</u> | <u>0.854</u> | 0.901  | 0.621                | 10.31 |
| ACI-EBB           | <b>0.183</b> | <b>0.458</b> | 0.915        | 1.423        | 1.562        | 0.908        | 0.913  | 0.665                | 19.55 |
| EBB (rule)        | 0.183        | 0.458        | 0.910        | 1.319        | 1.400        | 0.854        | 0.901  | 0.621                | 10.31 |

differences are systematic but heavy-tailed); on Carparts neither EBB nor TweedieGP separates from the other (adjusted  $p \geq 0.50$  both ways), consistent with the half-percent aggregate margin between them; and on Online Retail TweedieGP is ahead by 0.04% in mean SPL with the paired  $t$  at  $p = 0.71$ . EBB is significantly better than TweedieGP on Auto and RAF.

The paired MAE tests support a systematic improvement over the stationary hierarchy. Against TSB, tuned TSB, ADIDA, and IMAPA, mean MAE favors EBB but the adjusted paired  $t$ -tests are not significant; the Wilcoxon tests reverse because those methods win on the typical series while EBB reduces the difficult-series tail. On per-series RMSE, both adjusted tests favor EBB against every comparator at  $p < 0.05$ . Raw intervals mildly over-cover, most strongly on Carparts.

## E.6 Selection objective and efficiency

The selector is not mathematically tied to pinball loss; it scores the criterion supplied on the chronological validation tail. We use item-averaged SPL because the target is a predictive distribution. The best candidates are locally close, and a full cross-objective regret matrix remains future work.

Joint selection takes 51–243 seconds across the five panels, excluding data loading, and covers 24 forecast candidates with one deterministic EM/BIC search for the mixture on the fitting head. Final estimation is analytic apart from low-dimensional group optimization; selection and point prediction are deterministic, while probabilistic bootstrap averaging uses the released fixed seed.

## E.7 Measured cost of every method

Table 26 reports instrumented wall-clock for every method of Table 2 on Carparts (2,509 series), measured on one CPU workstation in a single session with configurations unchanged. The EBB row includes its complete 24-candidate joint selection in addition to the final fit ( $B=20$ ) and five-quantile prediction; the TSB-tuned row likewise includes its 25-candidate selection. No method uses a GPU. Figure 10 plots cost against mean SPL.

On the daily panels the official TweedieGP implementation was run in full under the authors’ released defaults (sparse variational GP with 200 inducing points and log-spaced initialization for series longer than 200 observations, all training points otherwise; Adam with learning rate 0.1; 50,000 predictive samples; median-demand scaling; restart on numerical failure) and the authors’ daily period. Online Retail (3,649 series) completed with no failed fits at a per-series median of 5.1 s; M5 (5,000 series, 1,294 training observations)

Table 12: Full Carparts walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov<sup>+</sup>@80) coverage and average interval width (AIW). Every baseline refits each of the six monthly blocks; EBB updates sufficient statistics. The zero row is excluded from ranking.

| Model             | q10          | q25          | q50          | q75          | q90          | Mean         | Cov@80 | Cov <sup>+</sup> @80 | AIW  |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------|----------------------|------|
| AutoARIMA         | 0.150        | 0.282        | 0.537        | 0.643        | 0.484        | 0.419        | 0.886  | 0.620                | 1.63 |
| AutoTheta         | 0.120        | 0.253        | 0.523        | 0.657        | 0.474        | 0.406        | 0.902  | 0.678                | 1.80 |
| iETS              | 0.272        | 0.368        | 0.493        | 0.532        | 0.515        | 0.436        | 0.442  | 0.411                | 1.17 |
| CP-Croston        | 0.099        | 0.245        | 0.480        | 0.614        | 0.539        | 0.395        | 0.847  | 0.580                | 1.39 |
| CP-SBA            | 0.098        | 0.242        | 0.475        | 0.611        | 0.538        | 0.393        | 0.847  | 0.581                | 1.37 |
| CP-TSB            | 0.141        | 0.301        | 0.487        | 0.513        | 0.476        | 0.384        | 0.834  | 0.551                | 1.27 |
| CP-ADIDA          | 0.096        | 0.230        | 0.411        | 0.529        | 0.488        | 0.351        | 0.829  | 0.593                | 1.28 |
| CP-IMAPA          | 0.101        | 0.235        | 0.413        | 0.518        | 0.479        | 0.349        | 0.830  | 0.581                | 1.27 |
| ACI-ADIDA         | 0.088        | 0.221        | 0.406        | 0.517        | 0.477        | 0.342        | 0.870  | 0.604                | 1.38 |
| TweedieGP         | 0.075        | 0.188        | 0.375        | 0.490        | <b>0.433</b> | <b>0.312</b> | 0.931  | 0.662                | 1.46 |
| Zero (diagnostic) | 0.075        | 0.187        | 0.374        | 0.560        | 0.672        | 0.374        | 0.795  | 0.000                | 0.00 |
| EBB               | <u>0.075</u> | <u>0.187</u> | <u>0.372</u> | <u>0.481</u> | 0.458        | 0.315        | 0.932  | 0.669                | 1.42 |
| ACI-EBB           | <b>0.075</b> | <b>0.187</b> | <b>0.371</b> | <b>0.480</b> | <u>0.452</u> | <u>0.313</u> | 0.934  | 0.676                | 1.41 |
| EBB (rule)        | 0.075        | 0.187        | 0.372        | 0.481        | 0.458        | 0.315        | 0.932  | 0.669                | 1.42 |

completed with no failed fits at a per-series median of 17.3 s, of which roughly half is the 50,000-sample predictive draw over the 647-step external span (the original protocol forecasts 28 steps), a protocol difference rather than a method cost. Two caveats govern every timing in Table 26. First, all rows were measured on one machine in one session, and absolute values are not comparable across platforms. Second, Damato et al. [6] report 0.24 s per Carparts series on an M3 MacBook Pro; on our machine the released defaults give 1.89 s (single thread, median of twelve series) and the paper-stated budget of 100 iterations gives 0.97 s, because the released code doubles the iteration cap for Tweedie likelihoods on series with at most 500 observations.

## E.8 Controlled-study diagnostics

In the length sweep used to isolate credibility leverage, the generator holds  $K = 2$ , 500 items per group, occurrence probabilities 0.05 and 0.80, log-size centers three units apart, and  $w = 1$ . The true partition’s pinball advantage falls monotonically from 1.53% at  $T = 12$  (median 4.5 positives per item) to 0.01% at  $T = 400$  (median 154), with NLL, Brier, and log-size MSE following the same direction. Across the main grid, the RMSE of  $\hat{\tau}_g^2$  approximately doubles as  $w$  falls from 1.00 to 0.90 ( $0.037 \rightarrow 0.144$  at  $T = 120$  and  $0.074 \rightarrow 0.149$  at  $T = 30$ ), directly measuring the resolution cost in Proposition 2.

The generator centers the groups symmetrically around a fixed grand mean so that increasing separation does not change total demand scale, and every arm is evaluated through the same hurdle-lognormal predictive representation under both likelihood-based proper scores and point metrics. A four-arm ladder—single pool, true labels with estimated hyperparameters, true labels with true hyperparameters, and the true conditional law—checks the implementation; the true conditional law is best on every metric in every cell.

**Separation axis.** The grid above pools the three separations  $\{0.5, 1, 2\}$  of the original sweep, and the length sweep uses a single separation of three; the envelope of Figure 5 therefore mixes separations. To resolve them we rerun the same generator on a pre-registered grid: separation  $\in \{0, 0.25, 0.5, 1, 2, 3\}$  log-units between adjacent group centers,  $w \in \{1, 0.95, 0.90\}$ ,  $T \in \{30, 120\}$ , nine paired replicates per cell (seed 20260913), four groups of sixty items,  $\tau^2 = 0.15$ ,  $\sigma^2 = 1$ , occurrence spread in logit scaled with the size separation as  $0.8 \min(\text{sep}, 1)$  so that zero separation removes structure from both blocks (a first pass held it fixed at 0.8; it is archived and superseded). Each cell records the true partition’s paired pinball gain over a single pool, the gain of the paper’s mixture objective run on the same panel, the median size-block credibility under the single pool, and the share of item log-size variance explained by the true labels ( $R^2$ , computed from the discounted item statistics so it includes sampling noise). Two remedies for the learned mixture that already exist in the released code are run as extra arms on the same panels: label sample-splitting (labels from one

Table 13: Full Auto walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov<sup>+</sup>@80) coverage and average interval width (AIW). Every baseline refits each of the six monthly blocks; EBB updates sufficient statistics. The zero row is excluded from ranking.

| Model             | q10          | q25          | q50          | q75          | q90          | Mean         | Cov@80 | Cov <sup>+</sup> @80 | AIW  |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------|----------------------|------|
| AutoARIMA         | 0.141        | 0.313        | 0.447        | 0.405        | 0.263        | 0.314        | 0.778  | 0.799                | 8.73 |
| AutoTheta         | 0.152        | 0.320        | 0.460        | 0.418        | 0.262        | 0.322        | 0.780  | 0.825                | 9.64 |
| iETS              | 0.172        | 0.312        | 0.435        | 0.423        | 0.301        | 0.329        | 0.564  | 0.690                | 7.33 |
| CP-Croston        | 0.148        | 0.307        | 0.437        | 0.402        | 0.296        | 0.318        | 0.801  | 0.802                | 5.97 |
| CP-SBA            | 0.146        | 0.304        | 0.434        | 0.404        | 0.301        | 0.318        | 0.800  | 0.803                | 6.01 |
| CP-TSB            | 0.155        | 0.321        | 0.458        | 0.414        | 0.301        | 0.330        | 0.808  | 0.799                | 5.97 |
| CP-ADIDA          | 0.149        | 0.307        | 0.437        | 0.400        | 0.295        | 0.318        | 0.802  | 0.802                | 5.92 |
| CP-IMAPA          | 0.149        | 0.307        | 0.437        | 0.400        | 0.295        | 0.318        | 0.802  | 0.802                | 5.92 |
| ACI-ADIDA         | 0.139        | 0.305        | 0.438        | 0.401        | 0.290        | 0.315        | 0.830  | 0.822                | 6.74 |
| TweedieGP         | 0.122        | 0.279        | <u>0.418</u> | <b>0.389</b> | <b>0.254</b> | 0.292        | 0.834  | 0.828                | 8.81 |
| Zero (diagnostic) | 0.112        | 0.281        | 0.561        | 0.842        | 1.010        | 0.561        | 0.262  | 0.000                | 0.00 |
| EBB               | <b>0.112</b> | <u>0.275</u> | 0.418        | 0.393        | 0.256        | <u>0.291</u> | 0.906  | 0.873                | 9.17 |
| ACI-EBB           | <u>0.113</u> | <b>0.275</b> | <b>0.416</b> | <u>0.391</u> | <u>0.256</u> | <b>0.290</b> | 0.905  | 0.876                | 9.41 |
| EBB (rule)        | 0.113        | 0.275        | 0.416        | 0.391        | 0.256        | 0.290        | 0.905  | 0.876                | 9.41 |

interleaved half, hyperparameters from the other) and credibility shrinkage of the group variance component, alone and combined. Table 28 summarizes the surface by separation and Table 29 reads it at the real panels’ coordinates. Real-panel  $R^2$  uses the same statistic under the learned mixture labels at the selected discount and under the ADI/CV<sup>2</sup> taxonomy; we do not use the learned  $\hat{\tau}_g^2$  as a scale because it collapses to its floor under learned partitions (Section 5.6). The simulation was run and summarized before the real panels were placed.

## E.9 Released tables and artifacts

Complete per-dataset tables with RMSE, WRMSSE, all quantile levels, coverage, interval widths, fixed-corner ablations, and runtimes accompany the code release, together with the selection surfaces and per-run diagnostics behind every table in this paper.

## E.10 Machine-readable significance output

The complete per-series results behind Table 24 and Table 21, including the RMSE tests and the Holm-adjusted families, accompany the code release as machine-readable CSV files.

## E.11 MASE

For completeness we report MASE (per-series MAE scaled by in-sample naive MAE), the intermittent-demand literature’s traditional point metric. It inherits the absolute-error pathology quantified in Section 5.2: the all-zero forecast attains the best MASE on Online Retail (1.84 vs. TSB 2.08, EBB 2.41), Carparts, and RAF, and zero-leaning methods (TSB, IMAPA) lead among real forecasters. On Auto, EBB (0.889) is second only to CrostonSBA (0.883). Full table in the repository (`mase_table.csv`). We treat MASE as descriptive, not decision-relevant, for the same reason as MAE.

## E.12 Walk-forward artifacts

The complete point and probabilistic results are in Tables 10 and 3; released block-level artifacts retain the seven-day boundaries and every per-block prediction.

Table 14: Full RAF walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov<sup>+</sup>@80) coverage and average interval width (AIW). Every baseline refits each of the twelve monthly blocks; EBB updates sufficient statistics. The zero row is excluded from ranking.

| Model             | q10          | q25          | q50          | q75          | q90          | Mean         | Cov@80 | Cov <sup>+</sup> @80 | AIW   |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------|----------------------|-------|
| AutoARIMA         | 0.066        | 0.151        | 0.490        | 0.742        | 0.592        | 0.408        | 0.944  | 0.352                | 8.76  |
| AutoTheta         | 0.066        | 0.158        | 0.497        | 0.809        | 0.623        | 0.431        | 0.948  | 0.409                | 10.03 |
| iETS              | 0.220        | 0.270        | 0.353        | 0.435        | 0.490        | 0.354        | 0.495  | 0.019                | 0.53  |
| CP-Croston        | 0.090        | 0.216        | 0.381        | 0.471        | 0.492        | 0.330        | 0.828  | 0.045                | 1.03  |
| CP-SBA            | 0.088        | 0.212        | 0.375        | 0.468        | 0.491        | 0.327        | 0.828  | 0.041                | 0.98  |
| CP-TSB            | 0.160        | 0.317        | 0.462        | 0.488        | 0.494        | 0.384        | 0.836  | 0.025                | 0.37  |
| CP-ADIDA          | 0.078        | 0.187        | 0.343        | 0.449        | <u>0.489</u> | 0.309        | 0.823  | 0.021                | 0.72  |
| CP-IMAPA          | 0.082        | 0.192        | 0.348        | 0.452        | 0.490        | 0.313        | 0.824  | 0.019                | 0.69  |
| ACI-ADIDA         | 0.057        | 0.154        | 0.326        | 0.446        | 0.560        | 0.308        | 0.909  | 0.083                | 1.85  |
| TweedieGP         | 0.053        | 0.134        | 0.270        | 0.417        | 0.526        | 0.280        | 0.935  | 0.198                | 4.07  |
| Zero (diagnostic) | 0.053        | 0.134        | 0.267        | 0.400        | 0.480        | 0.267        | 0.919  | 0.000                | 0.00  |
| EBB               | <u>0.053</u> | <u>0.134</u> | <u>0.267</u> | <b>0.400</b> | <b>0.486</b> | <b>0.268</b> | 0.922  | 0.041                | 0.95  |
| ACI-EBB           | <b>0.053</b> | <b>0.134</b> | <b>0.267</b> | <u>0.401</u> | 0.517        | <u>0.274</u> | 0.927  | 0.101                | 2.07  |
| EBB (rule)        | 0.053        | 0.134        | 0.267        | 0.400        | 0.486        | 0.268        | 0.922  | 0.041                | 0.95  |

Table 15: Full M5 walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov<sup>+</sup>@80) coverage and average interval width (AIW). Static CP wrappers and ACI-ADIDA refit each of the 93 seven-day blocks on the 5,000-series sample; AutoARIMA, AutoTheta, iETS, and TweedieGP were not run under this protocol on cost grounds; EBB updates sufficient statistics each block. The zero row is excluded from ranking.

| Model             | q10          | q25          | q50          | q75          | q90          | Mean         | Cov@80 | Cov <sup>+</sup> @80 | AIW  |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------|----------------------|------|
| CP-Croston        | 0.502        | 1.150        | 1.917        | 2.343        | 1.923        | 1.567        | 0.812  | 0.679                | 2.26 |
| CP-SBA            | 0.495        | 1.137        | 1.909        | 2.385        | 1.956        | 1.576        | 0.812  | 0.684                | 2.28 |
| CP-TSB            | 0.531        | 1.263        | 2.094        | 2.377        | 2.042        | 1.662        | 0.831  | 0.685                | 2.37 |
| CP-ADIDA          | 0.479        | 1.108        | 1.889        | 2.330        | 1.963        | 1.554        | 0.827  | 0.690                | 2.28 |
| CP-IMAPA          | 0.480        | 1.110        | 1.889        | 2.301        | 1.949        | 1.546        | 0.828  | 0.691                | 2.28 |
| ACI-ADIDA         | 0.402        | <u>0.960</u> | <b>1.718</b> | <u>2.211</u> | <u>1.806</u> | <u>1.419</u> | 0.883  | 0.718                | 3.48 |
| Zero (diagnostic) | 0.392        | 0.980        | 1.959        | 2.938        | 3.526        | 1.959        | 0.606  | 0.000                | 0.00 |
| EBB               | <u>0.391</u> | 0.967        | 1.852        | 2.336        | 1.880        | 1.485        | 0.875  | 0.690                | 2.77 |
| ACI-EBB           | <b>0.389</b> | <b>0.951</b> | <u>1.768</u> | <b>2.140</b> | <b>1.644</b> | <b>1.379</b> | 0.891  | 0.730                | 3.00 |
| EBB (rule)        | 0.389        | 0.951        | 1.768        | 2.140        | 1.644        | 1.379        | 0.891  | 0.730                | 3.00 |

## E.13 Neural baselines

Table 30 retains the earlier neural comparisons. Deep Renewal Processes (DRP) [27] are architecturally close to our occurrence-size decomposition, but the legacy subsample numbers are indicative only: we rebuilt the Python 3.8/MXNet 1.7 environment and launched the original code path, yet three configurations diverged to NaN between epochs 7 and 11 and failed during prediction. DeepAR and an RNN under prediction-only chaining reach lower MAE in one case but worse RMSE and RMSSE at both horizons. Neural decoding and a constant sufficient-statistic forecast are not timing-equivalent, so these rows should not support a broad efficiency ratio.

Two newly executed reference points provide a cleaner, narrower check. DeepState—the published Amazon ISSM successor available in GluonTS, not a reproduction of Seeger et al. [20]—obtains mean SPL 1.0652 on a 1,000-series Online Retail subset and 0.4608 on full Carparts. Chronos-Bolt-small obtains 0.8448 on an Online Retail 500-series subset, versus 0.8342 for EBB on exactly the same subset: a small 1.3% gap. Its Carparts result is only a reference point because the same-source Monash panel appears in the Chronos data repository and potential training contamination cannot be excluded. These subset results motivate an efficiency, determinism, and transparency claim, not a large foundation- model accuracy advantage.

Table 16: Per-panel mean-SPL regret (percent) to the best non-degenerate method on that panel, under the fixed-origin and walk-forward protocols. Zero regret marks the panel leader. Dashes mark methods not run under that protocol on that panel.

| Method      | Fixed origin |          |      |      |       | Walk-forward |          |      |      |      |
|-------------|--------------|----------|------|------|-------|--------------|----------|------|------|------|
|             | OR           | Carparts | Auto | RAF  | M5    | OR           | Carparts | Auto | RAF  | M5   |
| AutoARIMA   | 13.8         | 27.1     | 8.6  | 50.0 | 10.4  | 24.6         | 34.4     | 8.2  | 52.3 | —    |
| AutoTheta   | 13.7         | 22.1     | 15.9 | 53.1 | 10.6  | 24.2         | 30.0     | 11.1 | 60.7 | —    |
| Tweedie-GLM | 29.8         | 48.6     | 9.7  | 16.5 | 150.8 | —            | —        | —    | —    | —    |
| iETS        | 11.3         | 41.9     | 17.4 | 30.4 | 25.0  | 30.1         | 39.8     | 13.3 | 31.9 | —    |
| CP-Croston  | 75.3         | 24.8     | 11.0 | 24.0 | 4.2   | 60.9         | 26.7     | 9.7  | 23.1 | 13.7 |
| CP-SBA      | 75.1         | 23.9     | 10.9 | 22.8 | 4.4   | 61.5         | 25.9     | 9.5  | 21.9 | 14.4 |
| CP-TSB      | 32.8         | 17.7     | 17.8 | 42.3 | 15.8  | 61.1         | 22.9     | 13.7 | 43.3 | 20.5 |
| CP-ADIDA    | 27.1         | 11.1     | 11.5 | 15.6 | 5.7   | 35.8         | 12.4     | 9.4  | 15.4 | 12.7 |
| CP-IMAPA    | 27.5         | 10.8     | 11.6 | 16.3 | 5.9   | 38.1         | 11.9     | 9.5  | 16.6 | 12.1 |
| ACI-ADIDA   | —            | —        | —    | —    | —     | 6.8          | 9.5      | 8.5  | 15.1 | 3.0  |
| TweedieGP   | 0.0          | 0.0      | 1.6  | 3.1  | 0.0   | 0.0          | 0.0      | 0.8  | 4.5  | —    |
| EBB         | 0.0          | 0.5      | 0.0  | 0.0  | 3.0   | 0.2          | 0.8      | 0.3  | 0.0  | 7.7  |
| ACI-EBB     | —            | —        | —    | —    | —     | 6.5          | 0.2      | 0.0  | 2.3  | 0.0  |
| EBB (rule)  | —            | —        | —    | —    | —     | 0.2          | 0.8      | 0.0  | 0.0  | 0.0  |

## E.14 Far-tail quantiles on the sparsest panels

Scaled pinball at  $q \in \{0.95, 0.975, 0.99\}$  on the two panels where the standard levels discriminate weakly, under the identical fixed protocol (EBB raw,  $B=20$ ; iETS not re-run at these levels). The zero forecast’s scaled pinball is *linear* in  $q$  (its pinball at level  $q$  is  $q\mathbb{E}[y/\text{scale}]$ ), so the far tail is exactly where its apparent advantage must collapse; the question the table answers is who inherits the lead.

Three readings follow. First, **the saturation breaks as expected**: on RAF the zero forecast, marginally ahead at  $q \leq 0.9$ , falls behind every intensity-bearing method from  $q=0.95$  on, and on Carparts it is last by a factor of two. Second, **tail leadership is model-specific**: official TweedieGP wins all three Carparts cells, while EBB is second in all three; on RAF, AutoARIMA wins q95, TweedieGP wins q97.5, and EBB wins q99 by a near tie. Third, this table deliberately narrows the claim: EBB’s most consistent advantage is aggregate SPL, not universal extreme-quantile dominance. The conformal wrappers still deteriorate in the RAF far tail, where their per-level calibration information is sparse.

## E.15 Full-posterior inference on the same hierarchy

To separate the inference method from the model, we paired plug-in EB with a full-posterior sampler at each discount. Each pair uses the same single global pool, shared size-variance specification, discounted sufficient statistics, seed-controlled 500-series Online Retail subsample, evaluation span, and scoring code. The sampler uses Metropolis-within-Gibbs for the Beta hyperparameters, four deterministic chains of 4,000 sweeps, 2,000 burn-in iterations, and thinning by four. Both a stationary version ( $w=1$ ) and a power-prior discounted version ( $w=0.98$ ) are tested. The richer EBB arm runs its joint selection separately on this subsample and is reported only as context, not as an inference-method contrast.

Two readings. First, *the empirical-Bayes approximation is not a distributional-accuracy bottleneck*: paired full-posterior inference changes mean SPL by 0.007%/0.019%, q90 SPL by 0.018%/0.048%, and RMSSE by 0.025%/0.085% at  $w = 1/0.98$ . MAE is somewhat more sensitive but still changes only 0.70%/1.42%. Second, forgetting helps MAE under both inference methods, so that finding is not an artifact of the EB plug-in. The maximum split- $\hat{R}$  across  $(\alpha, \beta, \mu_0, \tau^2, \sigma^2)$  is 1.0021, and the minimum effective sample size is 819 of 2,000 retained draws. What this experiment does *not* measure: the richer external state-space samplers [2, 20] remain unrun (Limitations, item 6); this conjugate single-pool Gibbs is far cheaper than those models and is a control for the inference method, not a stand-in for them. The plug-in EB path retains bit-reproducibility and requires no convergence diagnostics, but at this scale raw speed does not separate the two.

Table 17: Short-history scaling: initialization window truncated to its last  $L$  observations per series (evaluation span unchanged; SPL scale from the full window). Median credibility weights at the audited discount, mean SPL of EBB, TweedieGP, and CP-ADIDA, and EBB’s advantage over TweedieGP. Mixture labels are re-learned on the truncated window.

| Panel         | $L$  | median $n$ | $\lambda^{(o)}$ | $\lambda^{(+)}$ | EBB    | TweedieGP | CP-ADIDA | Adv. (%) |
|---------------|------|------------|-----------------|-----------------|--------|-----------|----------|----------|
| Online Retail | 14   | 14         | 0.823           | 0.405           | 0.8894 | 0.8920    | 1.0106   | +0.3     |
|               | 28   | 28         | 0.870           | 0.506           | 0.8862 | 0.8879    | 1.1519   | +0.2     |
|               | 56   | 56         | 0.893           | 0.546           | 0.8852 | 0.8859    | 1.1247   | +0.1     |
|               | full | 119        | 0.895           | 0.553           | 0.8849 | 0.8845    | 1.1239   | -0.0     |
| M5            | 28   | 28         | 0.910           | 0.812           | 1.6925 | 1.6626    | 1.7182   | -1.8     |
|               | 56   | 56         | 0.928           | 0.839           | 1.6645 | 1.6774    | 1.7082   | +0.8     |
|               | 112  | 112        | 0.930           | 0.844           | 1.6697 | 1.6558    | 1.6912   | -0.8     |
|               | 224  | 224        | 0.931           | 0.842           | 1.6637 | 1.5995    | 1.7092   | -4.0     |
|               | 448  | 448        | 0.931           | 0.842           | 1.6670 | 1.6063    | 1.7024   | -3.8     |
|               | full | 1294       | 0.931           | 0.841           | 1.6707 | 1.6228    | 1.7150   | -3.0     |
| Auto          | 6    | 6          | 0.278           | 0.867           | 0.3009 | 0.3221    | 0.3363   | +6.6     |
|               | 12   | 12         | 0.377           | 0.920           | 0.2948 | 0.3017    | 0.3317   | +2.3     |
|               | full | 18         | 0.421           | 0.940           | 0.2937 | 0.2985    | 0.3275   | +1.6     |
| Carparts      | 6    | 6          | 0.585           | 0.180           | 0.3199 | 0.3279    | 0.3858   | +2.4     |
|               | 12   | 12         | 0.560           | 0.231           | 0.3197 | 0.3173    | 0.3628   | -0.8     |
|               | 24   | 24         | 0.596           | 0.245           | 0.3206 | 0.3145    | 0.3585   | -1.9     |
|               | full | 45         | 0.628           | 0.224           | 0.3229 | 0.3214    | 0.3571   | -0.5     |
| RAF           | 6    | 6          | 0.037           | 0.000           | 0.2730 | 0.3079    | 0.4071   | +11.3    |
|               | 12   | 12         | 0.075           | 0.682           | 0.2696 | 0.2811    | 0.3506   | +4.1     |
|               | 24   | 24         | 0.135           | 0.790           | 0.2669 | 0.2788    | 0.3208   | +4.3     |
|               | 48   | 48         | 0.152           | 0.891           | 0.2674 | 0.2769    | 0.3141   | +3.4     |
|               | full | 72         | 0.106           | 0.917           | 0.2680 | 0.2763    | 0.3097   | +3.0     |

## E.16 Datasets

Online Retail: UCI 352; returns/zero-price removed, 99.5% quantity cap, daily aggregation densified with zeros. M5: Kaggle M5-Accuracy; seed-controlled 5,000-series sample. Both M5 protocols use a two-thirds initialization split; the learned mixture is selected at  $w = 0.95$  on all three audited seeds. Auto/RAF: gluon-ts `intermittent-datasets` branch. Carparts: Zenodo 4656022, series without missing values and with at least one zero (2,509). Monthly panels use the last- $h$  holdout of Damato et al. [6].

Table 18: Mean SPL of alternative positive-size predictive laws computed from the same fitted EBB posterior (single fit, no bootstrap averaging), against TweedieGP.

| Size law               | OR     | Carparts | Auto   | RAF    | M5     |
|------------------------|--------|----------|--------|--------|--------|
| log-normal (released)  | 0.8849 | 0.3230   | 0.2937 | 0.2681 | 1.6730 |
| log- $t$ , $\nu=10$    | 0.8849 | 0.3228   | 0.2936 | 0.2680 | 1.6735 |
| log- $t$ , $\nu=5$     | 0.8850 | 0.3226   | 0.2940 | 0.2678 | 1.6744 |
| Gamma (moment-matched) | 0.8878 | 0.3227   | 0.2937 | 0.2673 | 1.6735 |
| TweedieGP              | 0.8845 | 0.3214   | 0.2985 | 0.2763 | 1.6228 |

Table 19: Item-specific forgetting rates with shrinkage strength  $\kappa$  selected on the outer validation tail.  $w^*$  is the panel-wide discount; “share  $< w^*$ ” is the fraction of items that selected faster forgetting than the panel rate;  $\kappa=\infty$  and  $\kappa=0$  give the fully shrunk and unshrunk endpoints. External mean SPL,  $B=20$ .

| Panel         | $w^*$ | Scalar | $\kappa^*$ | Per-item | $\Delta$ (%) | share $< w^*$ | $\kappa=\infty$ | $\kappa=0$ | TweedieGP |
|---------------|-------|--------|------------|----------|--------------|---------------|-----------------|------------|-----------|
| Online Retail | 0.95  | 0.8849 | 10         | 0.8846   | -0.03        | 0.19          | 0.8849          | 0.8845     | 0.8845    |
| Carparts      | 0.9   | 0.3229 | $\infty$   | 0.3229   | +0.00        | 0.00          | 0.3229          | 0.3298     | 0.3214    |
| Auto          | 0.99  | 0.2937 | 30         | 0.2940   | +0.10        | 0.14          | 0.2942          | 0.2931     | 0.2985    |
| RAF           | 0.997 | 0.2680 | 100        | 0.2678   | -0.07        | 0.16          | 0.2680          | 0.2675     | 0.2763    |
| M5            | 0.95  | 1.6707 | 3          | 1.6515   | -1.15        | 0.49          | 1.6751          | 1.8155     | 1.6228    |

Table 20: Separate occurrence discount  $w^{(o)}$  with the size discount fixed at the selected  $w$ ;  $w^{(o)*}$  is chosen on the initialization window’s validation tail. External mean SPL ( $B=20$ ) for the tied and selected rates, the share of evaluation rows whose q90 forecast is exactly zero, and TweedieGP for reference.

| Panel         | $w$   | $w^{(o)*}$ | Tied   | Selected | $\Delta$ (%) | q90 = 0 share             | TweedieGP |
|---------------|-------|------------|--------|----------|--------------|---------------------------|-----------|
| Online Retail | 0.95  | tied       | 0.8849 | 0.8849   | +0.00        | 0.362 $\rightarrow$ 0.362 | 0.8845    |
| M5            | 0.95  | 0.98       | 1.6707 | 1.6683   | -0.14        | 0.206 $\rightarrow$ 0.233 | 1.6228    |
| Auto          | 0.99  | tied       | 0.2937 | 0.2937   | +0.00        | 0.000 $\rightarrow$ 0.000 | 0.2985    |
| Carparts      | 0.9   | 0.8        | 0.3229 | 0.3204   | -0.77        | 0.081 $\rightarrow$ 0.000 | 0.3214    |
| RAF           | 0.997 | tied       | 0.2680 | 0.2680   | +0.00        | 0.279 $\rightarrow$ 0.279 | 0.2763    |

Table 21: Per-series mean SPL paired tests: EBB versus every comparator on each panel, Holm-adjusted within panel (paired  $t$  and Wilcoxon signed-rank; the family includes TweedieGP on every panel). Cells count comparators by outcome at the 5% level; named entries are the comparators for which EBB is *not* significantly better.

| Panel         | Family | Sig. better | Sig. worse    | Undecided       |
|---------------|--------|-------------|---------------|-----------------|
| Online Retail | 10     | 9           | 0             | TweedieGP       |
| M5            | 10     | 5           | 1 (TweedieGP) | CP wrappers (4) |
| Auto          | 10     | 10          | 0             | —               |
| Carparts      | 10     | 9           | 0             | TweedieGP       |
| RAF           | 10     | 10          | 0             | —               |

Table 22: Walk-forward per-series paired tests, Holm-adjusted within each (panel, reference) family; a comparison is significant when both the paired  $t$  and the Wilcoxon adjusted  $p$  are below 0.05. Families contain every walk-forward method with stored per-series quantiles; TweedieGP is not in the M5 family and the static conformal wrappers are not in the Online Retail family.

| Panel         | Reference | Family | Sig. better | Sig. worse             | Undecided          |
|---------------|-----------|--------|-------------|------------------------|--------------------|
| Online Retail | EBB       | 7      | 6           | 0                      | TweedieGP          |
| Carparts      | EBB       | 11     | 9           | 0                      | ACI-EBB, TweedieGP |
| Auto          | EBB       | 11     | 9           | 0                      | ACI-EBB, TweedieGP |
| RAF           | EBB       | 11     | 11          | 0                      | —                  |
| M5            | EBB       | 7      | 5           | 2 (ACI-ADIDA, ACI-EBB) | —                  |
| Online Retail | ACI-EBB   | 7      | 4           | 2 (EBB, TweedieGP)     | ACI-ADIDA          |
| Carparts      | ACI-EBB   | 11     | 9           | 0                      | EBB, TweedieGP     |
| Auto          | ACI-EBB   | 11     | 10          | 0                      | TweedieGP          |
| RAF           | ACI-EBB   | 11     | 10          | 1 (EBB)                | —                  |
| M5            | ACI-EBB   | 7      | 7           | 0                      | —                  |

Table 23: Validation-tail rule for the calibration layer. On the 80/20 chronological split of the initialization window, EBB is initialized on the head (labels learned on the head only) and walked forward over the tail with and without the adaptive conformal layer; the layer is applied in deployment iff its tail scaled pinball ( $q \in \{.5, .75, .9\}$ ) is lower. External columns are the walk-forward means of Table 3; “matches” marks panels where the rule picks the ex-post better variant.

| Panel         | Tail blocks | Tail plain | Tail ACI | Choice | Ext. EBB | Ext. ACI-EBB | Matches |
|---------------|-------------|------------|----------|--------|----------|--------------|---------|
| Online Retail | 4           | 0.7382     | 0.7967   | plain  | 0.8540   | 0.9082       | yes     |
| M5            | 37          | 1.5686     | 1.4457   | ACI    | 1.4851   | 1.3785       | yes     |
| Auto          | 3           | 0.3313     | 0.3304   | ACI    | 0.2910   | 0.2901       | yes     |
| Carparts      | 9           | 0.8975     | 0.9061   | plain  | 0.3145   | 0.3128       | no      |
| RAF           | 14          | 0.9846     | 1.0081   | plain  | 0.2681   | 0.2743       | yes     |

Table 24: Paired per-series MAE tests on Online Retail: EBB versus each alternative. Negative mean differences favor EBB;  $p$ -values are Holm-adjusted.

| vs.                  | Mean diff | $p_t$        | $p_{\text{Wilcoxon}}$ |
|----------------------|-----------|--------------|-----------------------|
| Stationary hierarchy | −0.165    | $< 10^{-29}$ | $< 10^{-11}$          |
| TSB                  | −0.040    | 0.532        | $< 10^{-155}$         |
| TSB-tuned            | −0.090    | 0.086        | $< 10^{-72}$          |
| ADIDA                | −0.036    | 0.359        | $< 10^{-88}$          |
| IMAPA                | −0.049    | 0.281        | $< 10^{-97}$          |
| CrostonSBA           | −0.872    | $< 10^{-26}$ | $< 10^{-59}$          |
| AutoARIMA            | −0.891    | $< 10^{-14}$ | $< 10^{-12}$          |
| AutoTheta            | −2.644    | $< 10^{-81}$ | $< 10^{-104}$         |

Table 25: Raw EBB interval quality across panels at nominal 80% coverage. No post-hoc calibration is used.

|             | OR    | M5    | Auto  | Carparts | RAF   |
|-------------|-------|-------|-------|----------|-------|
| Coverage@80 | 0.892 | 0.846 | 0.907 | 0.931    | 0.921 |
| AIW@80      | 9.25  | 2.81  | 9.21  | 1.47     | 0.94  |

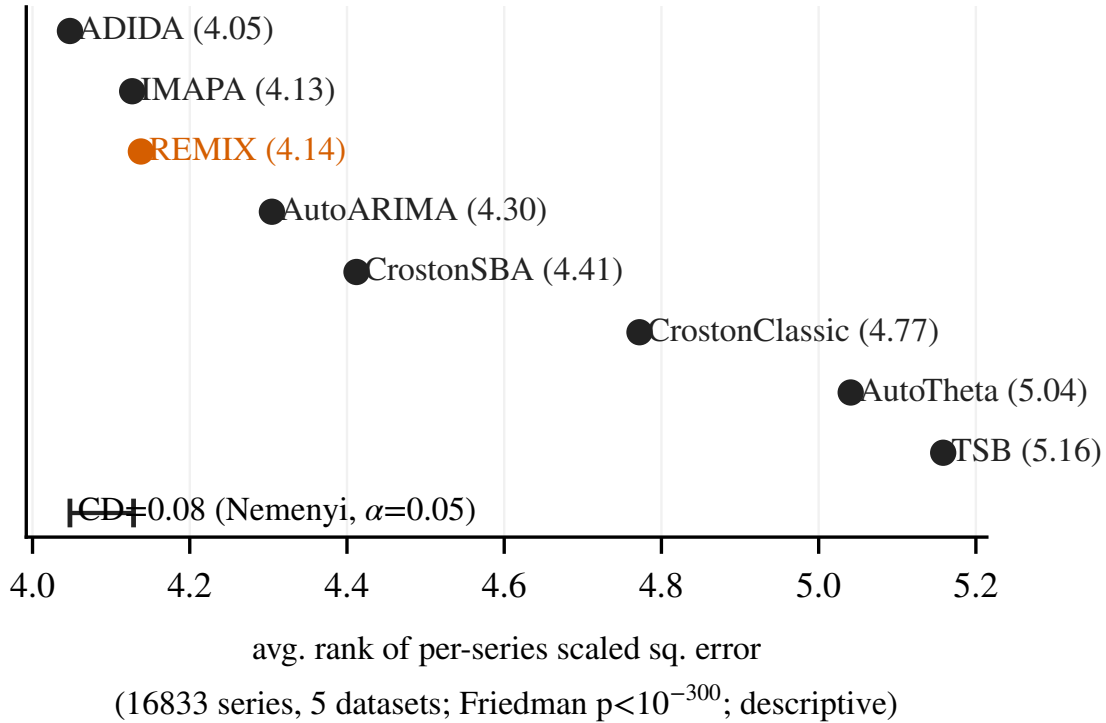


Figure 9: Average ranks of per-series scaled squared errors over the 16,833 series with complete saved row-level point predictions across the five panels. The critical-difference bar is shown only for scale: series, rather than datasets, are the blocks, so this is descriptive and not a valid Friedman–Nemenyi test.

Table 26: Instrumented wall-clock on Carparts, one machine, one session. “Sel.” marks totals that include the method’s own hyperparameter or candidate selection. AutoARIMA/AutoTheta share one StatsForecast call (split across the two); the five conformal wrappers share one residual-pool fit (split across the five). Rows below the rule are recorded from prior run metadata rather than re-timed: DeepState is a full-panel GluonTS/MXNet run, Chronos-Bolt is zero-shot inference on a 500-series subsample, and DRP is a 500-series Online Retail subsample (no Carparts run exists). All rows are CPU-only.

| Method                 | Total (s) | s/series | Sel. | Runtime      | Bit-repr. |
|------------------------|-----------|----------|------|--------------|-----------|
| EBB                    | 46.7      | 0.019    | yes  | Python       | Yes       |
| TSB-tuned              | 3.8       | 0.002    | yes  | Python       | No        |
| CP wrappers (each)     | 13.3      | 0.005    | no   | Python       | No        |
| Tweedie-GLM            | 37.3      | 0.015    | no   | Python       | No        |
| iETS                   | 55.3      | 0.022    | no   | R            | No        |
| AutoARIMA              | 56.2      | 0.022    | no   | Python       | No        |
| AutoTheta              | 56.2      | 0.022    | no   | Python       | No        |
| TweedieGP (12 workers) | 302.1     | 0.120    | no   | Python/torch | No        |
| DeepState              | 192.0     | 0.077    | no   | Python/MXNet | No        |
| Chronos-Bolt-small     | 8.3       | 0.017    | no   | Python/torch | No        |
| DRP (Deep Renewal)     | 1200.0    | 2.400    | no   | Python/MXNet | No        |

Table 27: Walk-forward wall-clock per method and panel (seconds unless marked in hours), covering every block of the protocol of Table 3: initialization, all per-block refits or updates, and prediction. StatsForecast, conformal, and ACI rows run single-process; TweedieGP runs its released code with twelve worker processes; EBB runs on one core and includes its initialization. AutoARIMA and AutoTheta share one call. <sup>†</sup>Projected from the measured 7,618 s full fit on the 5,000-series M5 sample at the Online Retail cadence of one refit per four blocks (24 refits); at one refit per block the projection is about 197 h. Neither was run.

| Method     | Online Retail | Carparts | Auto | RAF   | M5                          |
|------------|---------------|----------|------|-------|-----------------------------|
| AutoARIMA  | 1169          | 1.1 h    | 763  | 2.7 h | —                           |
| AutoTheta  | 1169          | 1.1 h    | 763  | 2.7 h | —                           |
| iETS       | 1795          | 270      | 213  | 1254  | —                           |
| CP-ADIDA   | —             | 9.6      | 4.4  | 38.0  | 465                         |
| ACI-ADIDA  | 126           | 7.4      | 8.9  | 27.3  | 581                         |
| TweedieGP  | 1.7 h         | 2686     | 2139 | 2.4 h | $\approx 51$ h <sup>†</sup> |
| EBB        | 22.4          | 3.4      | 3.6  | 13.5  | 36.6                        |
| ACI-EBB    | 52.3          | 4.2      | 4.8  | 13.3  | 217                         |
| EBB (rule) | 22.4          | 3.4      | 4.8  | 13.5  | 217                         |

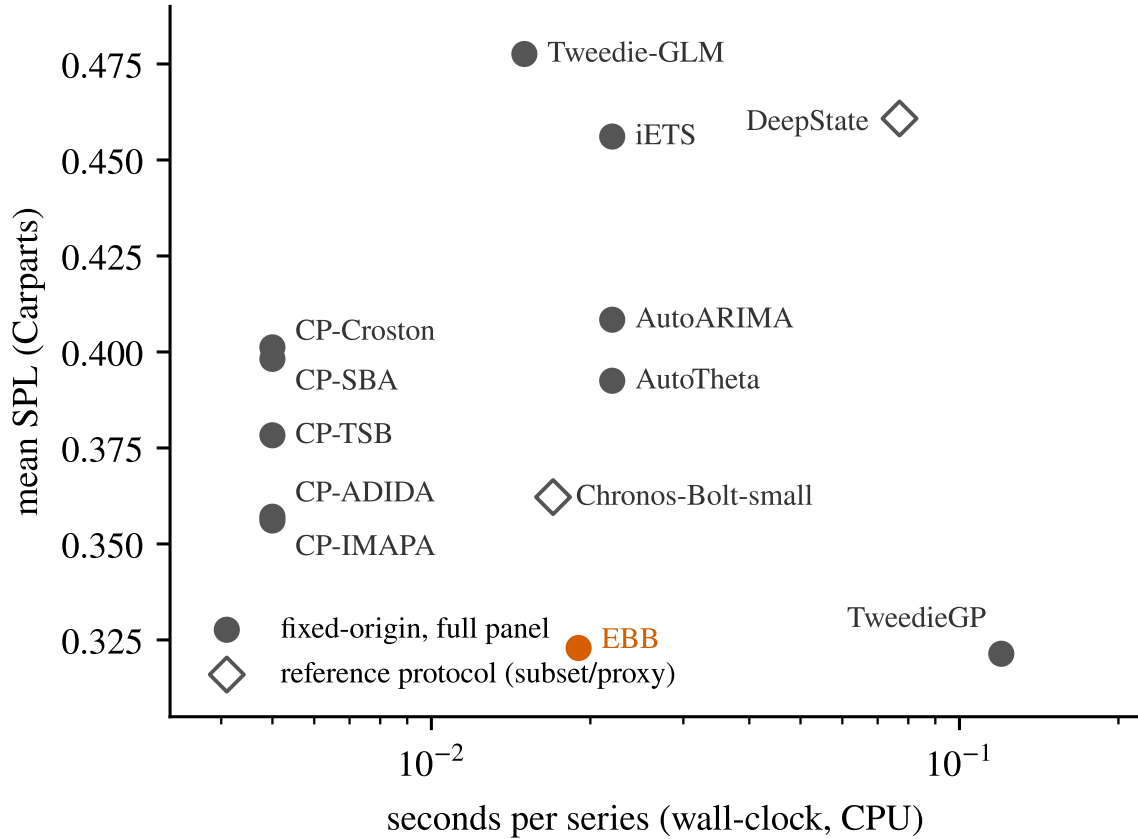


Figure 10: Wall-clock cost per series against mean SPL on Carparts. Filled circles are fixed-origin full-panel runs from Table 26; open diamonds are the reference points with subset or proxy protocols. All methods run on CPU.

Table 28: Separation-resolved controlled study, averaged over the six  $(w, T)$  cells at each separation: true-label  $R^2$ , the true partition’s gain over a single pool, the learned mixture’s gain, the same with label sample-splitting and credibility shrinkage combined, and the median number of components the mixture selects.

| Separation | $R^2$ | True (%) | Learned (%) | Learned+fix (%) | $K$ |
|------------|-------|----------|-------------|-----------------|-----|
| 0          | 0.01  | +0.07    | -0.02       | -0.36           | 1   |
| 0.25       | 0.19  | +0.12    | -0.01       | -0.30           | 1   |
| 0.5        | 0.47  | +0.70    | -0.92       | -0.94           | 2   |
| 1          | 0.75  | +0.20    | -1.47       | -0.86           | 2   |
| 2          | 0.93  | -0.08    | -0.71       | -0.37           | 3   |
| 3          | 0.97  | -0.30    | -0.37       | -0.51           | 4   |

Table 29: Real panels placed on the separation-resolved surface.  $w$  is the simulated discount matching the panel’s selected one;  $R^2$  is the size-block separation under the learned partition and under the taxonomy, and “family  $R^2$ ” the true-label  $R^2$  of the simulated family at the panel’s  $\lambda^{(+)}$  ( $T=120$ ). “Room” is the true partition’s gain interpolated along  $\lambda^{(+)}$  within the matching  $w$  at each length; “sim. learned” the learned mixture’s gain at the same point ( $T=120$ ); “realized” the panel’s internal refinement gain (Figure 5).

| Panel         | $w$  | $\lambda^{(+)}$ | $R^2$ learned | $R^2$ tax. | family $R^2$ | Room $T=30$ | Room $T=120$ | Sim. learned | Realized |
|---------------|------|-----------------|---------------|------------|--------------|-------------|--------------|--------------|----------|
| Online Retail | 0.95 | 0.553           | 0.44          | 0.17       | 0.63         | -0.52       | +0.63        | -0.87        | -0.05    |
| M5            | 0.95 | 0.841           | 0.79          | 0.30       | 0.86         | -0.34       | -0.45        | -1.21        | +0.58    |
| Auto          | 1    | 0.940           | 0.82          | 0.14       | 0.72         | -0.50       | -0.33        | -0.45        | -0.31    |
| Carparts      | 0.9  | 0.224           | 0.68          | 0.01       | 0.55         | +2.01       | +1.73        | -0.75        | -0.08    |
| RAF           | 1    | 0.917           | 0.95          | 0.09       | 0.62         | -0.48       | -0.41        | -0.58        | +0.03    |

Table 30: Neural comparisons on Online Retail. Top: probabilistic (unscaled pinball, 500-series subsample, fixed protocol). Bottom: point metrics under prediction-only chaining at two horizons.

|                 | q10   | q25   | q50   | q75   | q90   | Time    |
|-----------------|-------|-------|-------|-------|-------|---------|
| DeepRenewal     | 0.358 | 0.895 | 1.790 | 2.684 | 2.941 | ~20 min |
| Hierarchical EB | 0.387 | 0.914 | 1.772 | 2.390 | 2.186 | 1 s     |

|                 | MAE    |         | RMSE   |         | RMSSE  |         |
|-----------------|--------|---------|--------|---------|--------|---------|
|                 | $h=10$ | $h=100$ | $h=10$ | $h=100$ | $h=10$ | $h=100$ |
| RNN             | 3.257  | 3.132   | 11.938 | 12.538  | 0.875  | 8.197   |
| DeepAR          | 6.245  | 7.288   | 17.221 | 18.623  | 7.151  | 9.986   |
| Hierarchical EB | 4.639  | 4.495   | 11.284 | 11.733  | 0.855  | 8.170   |

| Model       | Carparts     |              |              | RAF          |              |              |
|-------------|--------------|--------------|--------------|--------------|--------------|--------------|
|             | q95          | q97.5        | q99          | q95          | q97.5        | q99          |
| CP-Croston  | 0.413        | 0.290        | 0.168        | 0.488        | 0.537        | 0.526        |
| CP-SBA      | 0.412        | 0.290        | 0.168        | 0.489        | 0.542        | 0.529        |
| CP-TSB      | 0.374        | 0.263        | 0.160        | 0.506        | 0.601        | 0.552        |
| CP-ADIDA    | 0.384        | 0.275        | 0.159        | 0.493        | 0.579        | 0.557        |
| CP-IMAPA    | 0.378        | 0.270        | 0.156        | 0.493        | 0.581        | 0.564        |
| AutoARIMA   | 0.385        | 0.332        | 0.301        | <b>0.460</b> | 0.395        | 0.356        |
| AutoTheta   | 0.367        | 0.316        | 0.285        | 0.464        | 0.399        | 0.359        |
| Tweedie-GLM | 0.581        | 0.445        | 0.305        | 0.497        | 0.439        | 0.367        |
| TweedieGP   | <b>0.355</b> | <b>0.222</b> | <b>0.128</b> | 0.462        | <b>0.352</b> | <u>0.247</u> |
| Zero        | 0.710        | 0.729        | 0.740        | 0.507        | 0.520        | 0.529        |
| EBB         | <u>0.365</u> | <u>0.251</u> | <u>0.144</u> | <u>0.461</u> | <u>0.364</u> | <b>0.246</b> |

|                        | SPL mean | SPL q90 | MAE    | RMSSE  | sec. |
|------------------------|----------|---------|--------|--------|------|
| EBB (incl. selection)  | 0.7942   | 1.3682  | 5.6259 | 4.6712 | 86.4 |
| EB global, $w=1$       | 0.7919   | 1.3588  | 5.5504 | 4.6684 | 0.2  |
| Gibbs global, $w=1$    | 0.7919   | 1.3590  | 5.5890 | 4.6696 | 2.7  |
| EB global, $w=0.98$    | 0.7935   | 1.3658  | 5.4774 | 4.6694 | 0.2  |
| Gibbs global, $w=0.98$ | 0.7937   | 1.3665  | 5.5549 | 4.6734 | 2.7  |